



Opportunities and challenges of noise interference suppression algorithms for dynamic ECG signals in wearable devices: A review

Juya Zhang^a, Yu Guo^{a,b}, Xinming Dong^{c,*}, Tong Wang^a, Jinhai Wang^{a,b}, Xin Ma^d, Huiquan Wang^{a,b,*}

^a School of Life Sciences, Tiangong University, Tianjin 300387, China

^b Tianjin Key Laboratory of Quality Control and Evaluation Technology for Medical Devices, Tianjin 300384, China

^c Tianjin Rehabilitation Convalescent Center, Tianjin 300191, China

^d School of Computer Science and Technology, Tiangong University, Tianjin 300387, China

ARTICLE INFO

Keywords:

Wearable devices
ECG signal
Noise suppression
Performance metrics

ABSTRACT

The primary obstacle in using wearable devices to record dynamic Electrocardiogram (ECG) signal for more efficient analysis of cardiac problems is noise interference, which can cause signal distortion and impact the accuracy of diagnosis and analysis. This paper examines noise suppression methods and their recent advancements in the time domain, frequency domain, time–frequency domain, and artificial intelligence domain, it assesses the various types of noise suppression, the advantages and disadvantages of the algorithms, real-time capabilities, noise reduction effectiveness, and performance metrics. This study also innovatively investigates the general applicability and specificity of the algorithms, discusses the shortcomings of the current ECG datasets, how to learn by analogy for noise reduction algorithms, the interpretability of the models as well as pointing out the problems and constraints of the existing algorithms for wearable devices as well as providing a unique perspective for future research directions.

1. Introduction

Cardiovascular diseases (CVDs) are a significant global health concern due to their high rates of illness, disability, and death. CVDs account for 27 % of global deaths and have emerged as the primary cause of death worldwide, presenting a major threat to human sustainable development [1]. In 2024, statistics reveal that in the United States, one person dies from a cardiovascular disease every 34 s, approximately 1,905 individuals die from heart disease daily, and there is roughly one heart attack every 40 s. In 2022 U.S. data, 72.1 % of adult out-of-hospital cardiac arrests happen at home or residence, with a survival rate to hospital discharge of 9.3 % for these non-traumatic instances [2]. An ECG is a frequently utilized test for diagnosing cardiovascular problems [3]. Continuous dynamic telecardiographic monitoring provides a more thorough monitoring of the patient's status and is superior to in-hospital surveillance. It can promptly gather data and implement life-saving actions when acute symptoms manifest. Due to significant advancements in chips, sensors, and artificial intelligence technologies, wearable devices are extensively utilized for patients with

cardiovascular disease who need ongoing health monitoring. These devices are designed to monitor patients during exercise, particularly when the heart is under increased stress. Conditions like coronary artery disease and myocardial ischemia may be more evident in the ECG signal. This results in a dynamic ECG signal that provides a more precise indication of the heart's health (Fig. 1).

Wearable ECG equipment monitors the patient's ECG signals, transmits them via WiFi or Bluetooth to a cell phone network, and facilitates data exchange over the Internet. The data will be uploaded to the remote ECG diagnostic cloud platform to form a data analysis ECG report, and connect to the remote ECG telemetry platform to realize real-time early warning of ECG monitoring. This allows doctors and medical monitoring centers to view the wearer's data in real time. In emergencies, EMS centers can dispatch ambulances promptly for first aid. (Create Independently).

Dynamic ECG signals recorded via wearable devices are highly susceptible to noise interference caused by physical activities and complex environments. This noise, often categorized as baseline wander, muscle artifact, electrode motion artifact, and industrial frequency interference,

* Corresponding authors at: Tiangong University, No. 399, Bingshui West Road, Xiqing District, Tianjin 300387, China.

E-mail addresses: xinming2024@126.com (X. Dong), huiquan85@126.com (H. Wang).

<https://doi.org/10.1016/j.measurement.2025.117067>

Received 16 May 2024; Received in revised form 18 February 2025; Accepted 19 February 2025

Available online 22 February 2025

0263-2241/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

significantly degrades the quality of ECG signals, directly impacting the accuracy of medical diagnoses. Current noise interference leads to the automatic exclusion of heavily corrupted signal segments, causing data loss and complicating the detection of potential diseases during motion. The need for portable and robust ECG monitoring systems is therefore paramount [4]. To address these challenges, the development of advanced noise suppression algorithms tailored to wearable devices is critical. Such algorithms should aim to enhance signal quality, ensure diagnostic accuracy, and minimize the risk of misdiagnosis in real-world scenarios.

Wearable ECG monitoring systems track the heart's electrical activity by measuring voltage differences between electrodes. During exercise, ECG readings are easily disrupted by various disturbances. The noises can be classified as extrinsic and intrinsic. Baseline wander and Industrial frequency interference are considered extrinsic noises, while Electrode motion artifact and Muscle artifact are considered intrinsic noises [5]. The characteristics of these four types of noises are detailed in Table 1. Aside from the four primary types of noise, other noises, including high-frequency sawtooth noise and Gaussian white noise, may be present in the ECG signal. These sounds are caused by high-frequency electromagnetic waves generated through coupling in the signal loop. To reduce or remove this form of noise, one can design hardware circuits strategically and implement filtering and smoothing techniques at both hardware and software levels. Muscle artifact, Baseline wander (below 5 Hz), and Electrode motion artifact (below 7 Hz) exhibit significantly higher levels during data collection compared to when the subject is at rest [6]. Fig. 2 illustrates the frequency range of the ECG signal and noise. Baseline wander predominantly impacts the low-frequency component of the ECG signal, while myoelectric interference disrupts the high frequencies with minimal amplitude. The motion artifact coincide with the distinctive wave frequency of the ECG QRS [7] and surpass the amplitude of the ECG signal. Motion artifact noise is the primary source of interference in wearable ECG monitoring, leading to variations of a few millivolts between electrodes and skin. Among the four noise types discussed, baseline wander, electrode motion artifact, and muscle artifact are illustrated in Fig. 2. These noise types were selected due to their significant impact on ECG signal quality and the challenges they pose for wearable systems. Industrial frequency interference, while relevant, is not depicted in the figure as it is relatively

straightforward to suppress with conventional notch filtering methods and thus less critical in the context of advanced noise suppression techniques discussed in this manuscript. Of particular note is the overlap between the frequency range of Electrode motion artifact (1–10 Hz) and the distinctive frequency band of the QRS complex (approximately 5–50 Hz). This overlap makes motion artifact noise especially challenging to suppress, as it can distort the amplitude and shape of the QRS complex, which is critical for cardiac diagnosis. This can blur details in the ECG data, distorting the signal and significantly impacting signal analysis [8].

This figure illustrates the frequency ranges of baseline wander, electrode motion artifact, and muscle artifact. Industrial frequency interference is not shown, as it is relatively easy to remove using standard notch filtering techniques and does not pose the same challenges as the other noise types.

2. Research methods

A method is required to effectively reduce the noise interference in the dynamic ECG obtained by the wearable device, as illustrated in Fig. 3. The enhanced quality and performance of wearable device components have minimized industrial frequency interference, which can be efficiently mitigated by utilizing a 50 Hz/60 Hz trap filter. Hence, it is crucial to effectively reduce the other three types of noise present in the dynamic ECG output.

Created by the authors to illustrate the proposed framework for noise suppression methods.

ECG signal processing methods can be categorized into time domain, frequency domain, and time–frequency domain, which represent different inherent characteristics of ECG signals. In addition, Artificial Intelligence (AI) has emerged as a powerful approach for noise determination and suppression. While AI is not a signal domain in the traditional sense, it leverages computational models to analyze and process signals for noise suppression. Fig. 4(a) details the methods within each domain, while Fig. 4(b) shows the distribution of articles across the domains from 2016 to 2024, totaling 78 articles. The analysis reveals a higher number of articles in the time domain and artificial intelligence domain. Fig. 4(c) displays the distribution of different approaches across domains, with wavelet transform being predominant in

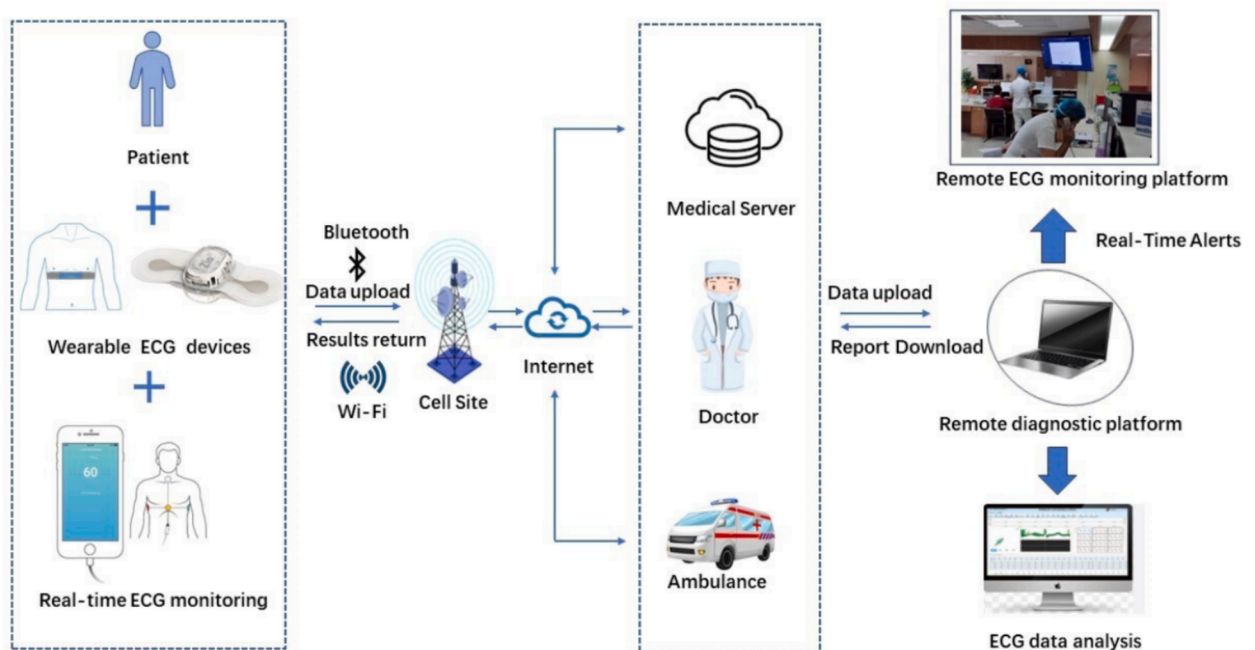


Fig. 1. Working mode of wearable ambulatory ECG device.

Table 1
Major noises in the ambulatory ECG.

Noise type	Reason	Frequency (H z)	Maximum Relative Amplitude (ECG)	Effect
Baseline wander [9]	Breathing, change of position or movement of electrode position	0.03–2	15 %	May result in a higher likelihood of myocardial infarction, Brugada syndrome, and other misdiagnoses due to ST-segment abnormalities
Industrial frequency interference [10]	Electromagnetic interference affecting the power supply and surroundings of cardiac testing equipment	50/60	50 %-200 %	Alters the magnitude, length, and form of weak, concentrated waves in the ECG signal
Muscle artifact [11]	High-frequency noise produced by human muscular contractions	5–1000	10 %	Modifying the morphology of localized ECG signal waves
Electrode motion artifact [12]	Displacement of the electrode–skin contact surface causes a change in the impedance of the skin electrodes	1–10	100 %-500 %	Overlap in frequency with ECG signals causes temporary alterations in the baseline

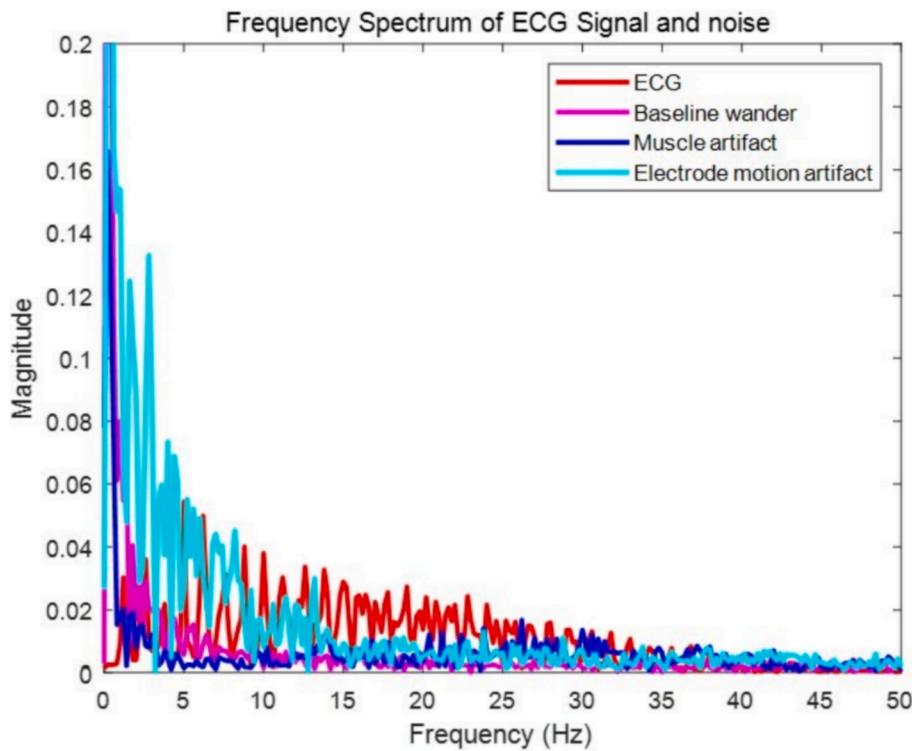


Fig. 2. Frequency Spectrum of ECG Signal and noise (Create Independently).

the time domain and deep learning models in the AI domain. This classification aids in considering various processing viewpoints and requirements for choosing specific processing methods, enabling more accurate customization of suppression and intervention to match the features of the ECG noise signal. Sections 2.1 to 2.4 offer an in-depth examination and overview of these methods in that order. The reviewed articles were selected based on a comprehensive search of databases such as PubMed, IEEE Xplore, and Scopus, covering publications from 2016 to 2024. The selection criteria focused on studies addressing ECG noise suppression in wearable devices, emphasizing advances in filtering methods, machine learning approaches, and hybrid techniques. Articles were included if they met the following criteria:

- 1) Published in peer-reviewed journals or conferences.
- 2) Focused on noise suppression techniques applicable to dynamic ECG signals.
- 3) Presented quantitative evaluations or theoretical advancements.
- 4) Duplicate and irrelevant articles were excluded based on title, abstract, and full-text screening.

2.1. Time-domain based noise suppression methods

2.1.1. Adaptive filtering methods

Under human motion conditions, adaptive filtering can effectively track the signal and suppress complicated motion artifact noise. These methods involve obtaining reference signals that have a strong connection with ECG signals or motion artifact. Among them, adaptive filtering methods based on the Least Mean Square (LMS) algorithm [13] and the Recursive Least Squares (RLS) algorithm [14] have been extensively researched. Adaptive filtering optimizes the effect by constantly updating the weights of the filter, as shown in Fig. 5, where ECG represents the ECG signal containing motion artifact, $d(n)$ serves as the reference signal, $\hat{d}(n)$ is the output signal, $e(n)$ is the error signal, which denotes the difference between the output signal and the reference signal, and $\Delta w(n)$ is the correction factor for the filter coefficients. The transfer function is corrected by feeding back the result of the error signal, and the correction factor is generated based on the input signal and the error signal.

The analysis was conducted on various reference signals. Adaptive filtering is necessary for rapid convergence when noise levels are high,

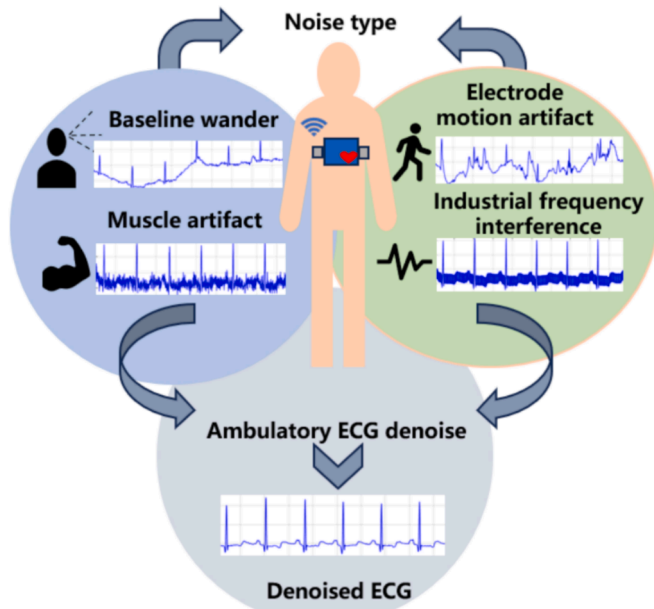


Fig. 3. Overall flowchart of ambulatory ECG noise suppression.

while high convergence accuracy is needed when noise levels are low. Existing adaptive filters face challenges in balancing convergence speed and accuracy. An adaptive forgetting factor-based filtering method solves this problem, as described in the reference [15]. It employs an acceleration sensor as a reference signal to achieve rapid convergence speed and high accuracy in the filtering process. In [16], Castano et al. simultaneously obtained ECG signals and three-axis motion data to create a reference signal using an autoregressive model. Similarly, An et al., in reference [17], used respiration-induced thoracic Impedance Pneumatic (IP) signals as reference signals. Impedance pneumography (IP) signals are used as a reference input for adaptive filtering in ECG

noise suppression. IP signals reflect the electrical impedance changes in the thorax due to respiratory activity and other bodily movements. Since both ECG and IP signals are recorded using the same electrodes, the IP signal captures motion artifacts that coincide with those present in the ECG signal. This strong correlation makes IP signals an effective reference for reducing motion artifacts in wearable ECG monitoring systems. Both methods effectively minimize motion artifact. Biopotential difference and electrode-tissue impedance signals from two auxiliary electrodes were utilized as reference signals in [18] and [19], surpassing the typical LMS method and successfully reducing high-intensity motion artifact. Furthermore, research has shown that the algorithm generates internal reference signals instead of obtaining extra ones. For example, the study in [20] employs an adaptive filter with three sequences and ECG and QRS removal techniques to create a reference signal. Similarly, the study in [21] integrates adaptive filtering with a smooth wavelet transform, utilizing motion artifacts obtained through threshold filtering as the reference signal.

Recent studies have indicated that triaxial acceleration signals are more prevalent than reference signals. Furthermore, triaxial motion signals are often combined with autoregressive models, IP signals, electrode-tissue impedance, and biopotential difference, which are commonly mentioned reference signals. IP signals and electrode-tissue impedance have high hardware demands for wearable devices. While research indicates that electrode-tissue impedance signals are better at eliminating motion artifact [22], implementing them on portable devices may be challenging due to hardware limitations. Hence, the selection of the reference signal to eliminate motion artifact should be determined according to particular requirements and real-world scenarios.

2.1.2. Traditional and adaptive filtering methods

Traditional filtering methods are not optimal for eliminating noise interference [23]. In contrast, Particle Filter (PF) is a helpful method for reducing noise and determining the state of a dynamic system. It can precisely estimate the location of the R-wave peak in the ECG signal for heart rate measurement [24]. A Monte Carlo approach is utilized in

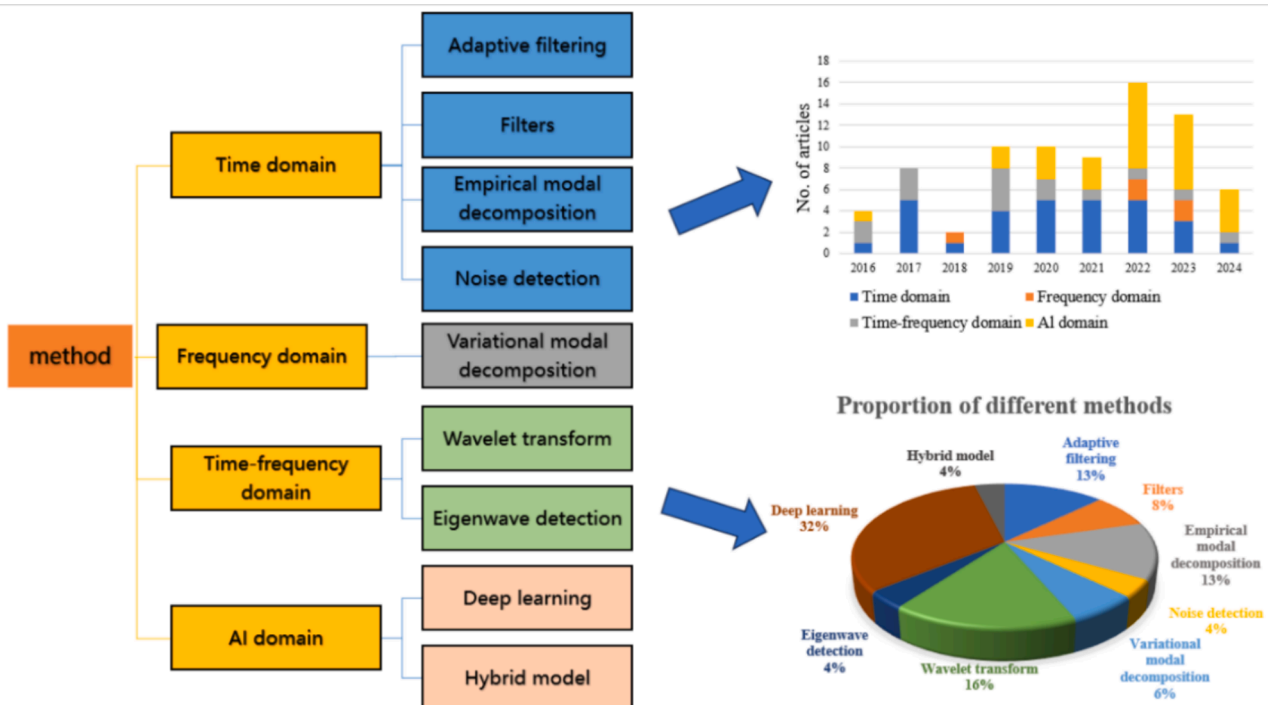


Fig. 4. (a) Classification of noise reduction methods for ECG signals (b) Statistics on the number of publications in the four major domains of methods from 2016 to 2024 (c) Statistics on the percentage of various methods under each domain (Create Independently).

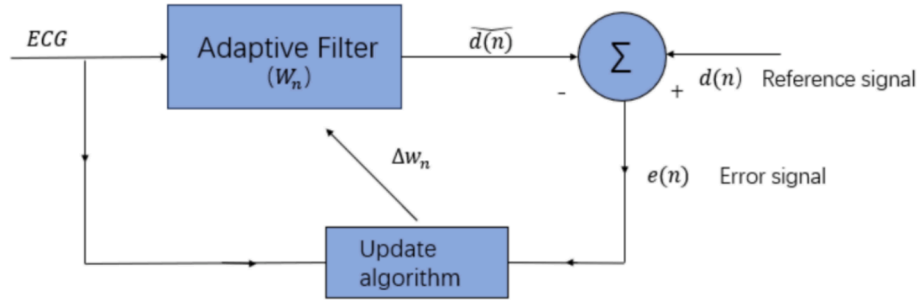


Fig. 5. Block diagram of adaptive filtering (Create Independently).

particle filtering to modify the frequency subband of the affected heartbeat and choose the most suitable wavelet type for reducing motion artifact with the assistance of a neural network [25]. A novel particle filtering algorithm was presented in reference [26], integrating the parameter γ , determined from the data of a six-axis motion sensor in a wearable device, to enable adaptive updates of particle weights. Monte Carlo Filtering (MCF) [27] provides a prediction, filtering, and smoothing method for dealing with nonlinear and non-Gaussian state-space problems, such as in [28], where the MCF technique is utilized to suppress motion artifact by introducing a new mechanism for updating the weights and predicting new values for the particles.

Model-based Bayesian frameworks are efficient in eliminating high-frequency interference, but they rely heavily on previous models derived from ECG signals. A Bayesian framework that includes dual adaptive Kalman Filtering (KF) was suggested to enhance adaptability to various signal patterns and reduce muscle artifact noise [29]. Introducing angular velocity as the third state of the kinetic model in the literature [30] is more efficient in enhancing the signal-to-noise ratio than the extended Kalman filter framework. Using the above method may cause signal distortion when the ECG signal experiences significant transitory fluctuations. In contrast, smoothing techniques like the Savitzky-Golay (SG) filter can decrease signal distortion while preserving key signal features like peak, amplitude, and width. Huang et al. [31] introduced an adaptive SG filtering approach that relies on discrete curvature estimation. This method retains the effective smoothing properties of the SG filter while also enhancing signal-to-noise ratio and signal fidelity. Recent advancements in ECG noise suppression have increasingly explored knowledge-based filtering and automatic noise type recognition. These approaches leverage domain-specific knowledge and machine learning to enhance filtering performance, offering promising results in reducing complex noise in wearable ECG systems. Although not the primary focus of this review, these techniques represent a significant direction for future research and development in the field of ECG signal processing.

2.1.3. Empirical modal decomposition methods

Empirical Mode Decomposition (EMD) [32] addresses the limitations of the wavelet transform by breaking down the signal into an Intrinsic Mode Function (IMF). Combining EMD, sample entropy, and an enhanced threshold function makes it possible to reconstruct and smooth ECG signals without the need to discard the first-order IMF directly [33]. Utilizing a period-delay approach, the combination of EMD and wavelet transform is employed to analyze the edges of the ECG signal. It decreases Baseline wander noise by enhancing the wavelet inverse transform [34]. However, EMD can result in the loss of edge information and modal mixing. An integrated Ensemble Empirical Modal Decomposition (EEMD) technique is suggested to enhance the situation by minimizing modal mixing [35]. This is achieved by introducing white noise to the data and leveraging its spectral characteristics. Yue et al. [36] used EEMD to decompose the ECG signal for the first time and decompose the highest frequency component for the second time, such as in [37] using modified empirical modal decomposition

combined with a Non-Local Means (NLM) for thresholding and reconstruction, which improves the problem of insufficient localized mean in the denoising of NLM, both methods maximize the preservation of information of the ECG signal, and at the same time suppressing high-frequency interference (both muscle artifact and Additive White Gaussian Noise (AWGN) belong to high-frequency interference). Zhang et al. [38] introduced the integrated EMD approach, which utilizes adaptive thresholding and peak filtering according to the 3σ criterion to reduce motion artifact and high-frequency interference efficiently. Studying different signal decomposition methods such as EMD, EEMD, Discrete Wavelet Transform (DWT), and Smooth Wavelet Transform (SWT), Sraith et al. [39] combined each method with a local mean (LM) filtering and found that the EEMD and LM methods performed best in suppressing high-frequency noise. The addition of auxiliary noise results in increased noise residuals in the IMF. To address this issue, the Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) technique was introduced [40], which inherits the advantages of EEMD and reduces the error in the reconstruction of ECG signals. For example, Xu et al. [41] utilized wavelet thresholding and CEEMDAN to suppress and rectify Baseline wander effectively. In view of the excellent performance of machine learning in signal denoising, in the literature [42], CEEMDAN is integrated with neural networks to prevent overfitting and effectively decrease low-frequency noise in ECG signals.

2.1.4. Noise detection methods

Various techniques are initially employed to precisely detect motion disturbances and subsequently execute specific removal procedures. Xie et al. [43] suggested detecting motion artifact by using domain expertise and adaptive noise thresholding to pinpoint potential artifact locations, which are then mitigated through local scaling and morphological filtering. A modified Complete Ensemble Empirical Mode Decomposition (CEEMD) method is applied in [44] to deconstruct the signal and identify the noise by analyzing parameters, including the maximum absolute amplitude of the autocorrelation function, the number of zero crossings, and the local maximum peaks in high and low frequencies. In literature [45], a RUSBoost classifier is trained to detect artifacts by utilizing features derived from the autocorrelation function. While it does not focus on eliminating artifacts, its detection method is nevertheless valuable.

2.2. Frequency domain based noise suppression methods

2.2.1. Variational modal decomposition methods

VMD is an adaptive signal decomposition method that iteratively decomposes a signal into a set of modes, each with a limited bandwidth and a defined center frequency. Although VMD is fundamentally an optimization-based approach, its operation heavily relies on frequency characteristics by extracting intrinsic modes that correspond to distinct spectral components. This aligns VMD with frequency-domain methods, as it effectively maps signal features to specific frequency bands. In recent research, VMD has shown superior performance in suppressing

baseline wander and muscle artifacts by isolating and reconstructing key frequency components [47–51]. The optimization process within VMD ensures precise frequency-based noise reduction, which complements traditional filtering techniques. While it may differ in implementation from classic frequency-domain approaches, its functionality and outcomes justify its inclusion in this category. Variational modal decomposition (VMD) is an optimization-based method that efficiently addresses the issue of modal aliasing in EMD by adaptively decomposing the signal into major modes (VMF) in a non-recursive manner [46]. Singh et al. [47] decomposed the ECG signal into VMFs to showcase VMD's outstanding effectiveness in reducing high-frequency interfering noise. They integrated the techniques of NLM and DWT to enhance the denoising outcome. Nevertheless, the conventional VMD method exhibits limited generalization capability, prompting researchers to enhance or integrate it. The choice of k and ∂ values in VMD had a substantial impact on the outcomes. Researchers used an adaptive particle swarm optimization technique to alter the parameters and enhance the program's performance [48]. In the literature [49], Malghan et al. utilized a grasshopper-based optimization algorithm to discover the parameters k and ∂ . They integrated this algorithm with dynamic temporal regularization to detect VMFs linked to high-frequency disturbances. In addition, when combining the VMD algorithms, a two-phase VMD method was proposed in [50] for enhancing the QRS wave-cluster features and facilitating the detection of the R-peak to improve the heart rate variability estimation and feature extraction efficiency. VMD, together with the Sparrow Search Algorithm and Singular Value Decomposition methods [51], helps reduce Baseline wander noise and maintain the morphological characteristics of the ECG signal. VMD is a robust technique that blends optimization and frequency-based signal analysis. While its adaptive decomposition process may distinguish it from conventional frequency-domain methods, the resulting modes are inherently tied to specific frequency bands, making it functionally similar to traditional approaches in this domain. Therefore, this manuscript categorizes VMD under frequency-domain methods to reflect its practical applications and outcomes in frequency-driven noise suppression. Future studies may explore VMD's hybrid nature and its potential integration with other domain-based methods.

2.3. Time-frequency domain based noise suppression methods

2.3.1. Wavelet transform methods

Conventional noise reduction techniques often include high and low-pass filters, which can lead to signal distortion when specific noise frequencies surpass the filter cutoff frequency [52]. The wavelet transform can reveal the time–frequency details of ECG signals [53]. It is crucial to choose a suitable threshold when applying the DWT to reduce noise in ECG signals. Using a generalized threshold may overly smooth the signals and diminish important features. Jenkal [54] and Gon et al. [55] introduced adaptive dual thresholding and enhanced S-median thresholding strategies to address the limitations of DWT threshold selection by modifying factor values for improved adaptivity. Baseline wander noise can be efficiently reduced by using power spectral density in [56]

and mathematical morphological filtering in [57]. The SWT is introduced to address the loss of information caused by DWT downsampling. SWT maintains a constant signal length and preserves the most valuable information. The SWT signal decomposition is shown in Fig. 6, a_c represents the approximation coefficients, which are the outputs of the low-pass filter ($H_l(w)$), and d_c represents the detail coefficients, which are the outputs of the output of the high-pass filter ($G_l(w)$), and this process continues up to the “n” decomposition layer [58].

Nagai et al. [59] employed the SWT technique to detect QRS wave clusters by the analysis of wavelet coefficient energy, but the method's robustness was low. Therefore, Berwal [60] et al. combined SWT with horizontal thresholding to reduce motion artifact, resulting in better performance compared to the method described in the literature [59]. The traditional wavelet transform is ineffective in capturing the edge details of the ECG signal and necessitates the choice of suitable wavelet functions and thresholds, hence restricting its adaptability.

The Empirical Wavelet Transform (EWT) was introduced to address these issues by deriving wavelet bases directly from the data to match the signal better, surpassing the constraints of the conventional wavelet transform that relies on fixed basis functions [61]. The EWT combined with WT maintains the features of the initial ECG QRS complex waveform and decreases motion artifact [62]. In addition, the EWT is initially used to process the ECG by translating the ECG signal to the frequency domain by the Fourier transform, as outlined in references [63] and [64]. Then the dominant component of the ECG signal is rebuilt to suppress Baseline wander noise. Yu et al. [65] employed an adaptive neural network based on signal estimation for processing, which outperforms classic wavelet thresholding methods like “Heursure” and “Minimax.”

2.3.2. Eigenwave detection methods

An inverse approach can be employed to detect and eliminate noise by concentrating on finding the essential components of the ECG signal. For example, Zou et al. [66] removed baseline offsets using cubic spline interpolation on the midpoints of the Q- and R-waves. They then shift-averaged the SQ segment to minimize noise between two QRS wave clusters, allowing for the reconstruction of the P- and T-waves. Agostinelli et al. [67] introduced a filtering technique that enhances ECG templates by segmenting the ECG signal into QRS and TUP segments. This method employs modulation and demodulation techniques for adaptive noise reduction, offering more excellent noise resistance compared to the traditional template method. These methods are primarily effective for normal rhythm ECG signals but may not be as effective for patients with arrhythmia. Therefore, Bashar et al. [68] examined long-term ECG signals impacted by motion artifact in an intensive care unit and successfully decreased the rate of false detections in disease diagnosis by differentiating ECG signals from noise artifacts using time and frequency domain characteristics.

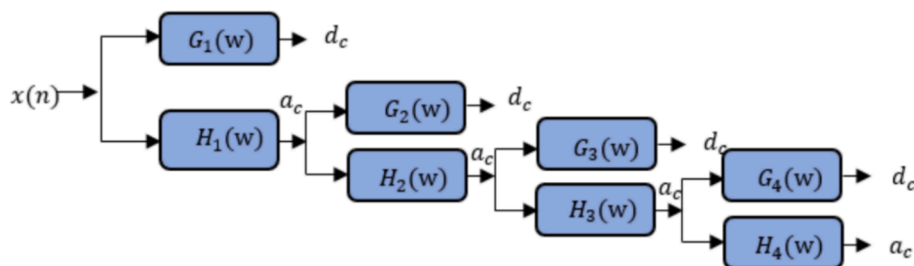


Fig. 6. SWT decomposition block diagram (. Adapted from Reference [58])

2.4. Noise suppression methods based on Artificial intelligence domain

2.4.1. Deep learning methods

Deep learning methods have demonstrated significant promise in reducing noise in ECG signals, with research concentrating on neural networks and noise reduction encoders. Denoising Auto-Encoder (DAE) [69] is an extension of the auto-encoder, where the input of DAE is a noisy ECG signal x , which is a combination of a clean ECG signal $\tilde{x}$ and a noisy signal y , and the output is a noise-canceled signal $\hat{x}$. The encoder maps the input signal x to a low-dimensional layer m through a neural network layer, and the decoder reconstructs the output signal $\hat{x}$ from the low-dimensional layer m . The framework is shown in Fig. 7.

For example, after converting the signal to the time–frequency domain, a sophisticated two-level network is applied for noise reduction [70]. The literature [71] demonstrates that combining recurrent neural networks (RNN) and deep neural networks (DNN) outperforms standard filtering strategies. This strategy primarily emphasizes enhancing the overall signal-to-noise ratio while overlooking the significance of peak preservation. However, this method mainly focuses on improving the overall signal-to-noise ratio and neglects the importance of peak preservation. To address this issue, the study in [72] improved this by effectively preserving the R-wave structure and utilizing DAE to achieve noise reduction without the need for an R-peak detection algorithm for signal alignment [73]. A further investigation utilized noise reduction by arranging the signals into a two-dimensional structure according to the correlation of cardiac cycles using a CNN model [74]. An Fully Convolutional Network (FCN) was presented for the first time to showcase better performance compared to DNN and CNN [75]. The two-stage U-net network proposed by Qiu et al. [76], as well as literature [77,78], and literature [79], which combine the Attention Mechanism and Transformer with the Convolutional Noise Reduction Auto-Encoder, respectively, can reduce waveform distortion. To improve the efficiency of the model, Jhang et al. [80] designed a low-memory connectivity noise-reducing self-encoder (LMSC-DAE) that maintains the key features at each layer through average pooling and point-by-point convolution [81]. The Lightweight U-shaped network design (LUNet) leverages specific network features to pinpoint noise location, reducing the chance of misdiagnosis and enabling partial signal discard [82]. Denoising Autoencoder (DAE) is an extension of autoencoders designed to reconstruct clean ECG signals from noisy inputs. The encoder maps the noisy input to a low-dimensional representation, and the decoder reconstructs the clean signal. Studies (e.g., Reference [69]) have demonstrated that DAEs significantly improve the signal-to-noise ratio (SNR) and reduce mean square error (MSE) in dynamic ECG noise suppression. Advantages: High adaptability to various noise types, such as baseline wander and motion artifacts. Challenges: Computationally intensive and requires a large amount of labeled data for training. With advancements in Generative Adversarial Network (GAN), researchers are now utilizing them in the cardiology discipline.

A GAN is an unsupervised deep learning model that comprises a Generator and a Discriminator, as seen in Fig. 8. The Generator aims to create data that closely resembles actual data, challenging the Discriminator to differentiate between the two. The Discriminator's objective is to accurately differentiate between input data from the real

dataset and that from the Generator. Both networks enhance their performance over time by ongoing iterative training until they reach a state of dynamic equilibrium.

Generative adversarial networks (GANs) have been employed in processing ECG signals to efficiently capture subtle features, as discussed in references [83–85]. The integration of GANs with LSTM layers has shown a significant enhancement in retaining medical information within ECG signals, as indicated in reference [86]. Wang et al. [87] employed conditional generative adversarial networks (CGANs) with noisy ECG signals to effectively preserve the signal's inherent characteristics. LUNet surpasses enhanced noise-reducing self-encoders [88] but is comparable to GAN [85]. However, CGANs and adversarial denoising convolutional neural networks (ADnCNN) [89] consistently outperform LUNet. Romero et al. [90] confirmed that multicore linear and nonlinear filtering methods can reduce Baseline wander noise effectively. A generative model for ECG noise reduction based on conditional fractional diffusion was proposed [91]. It can learn the mapping from noisy signals to clear signals and has better noise reduction performance compared to Romero et al.'s method, especially in handling extreme noise. GANs utilize a generator to create synthetic clean signals and a discriminator to distinguish between real and generated signals. This iterative training process allows GANs to model noise distributions effectively. References [85,87] highlight the superiority of GANs in handling composite noise scenarios, achieving better SNR and lower MSE compared to traditional methods. Conditional GANs (CGANs) have further improved the preservation of ECG signal characteristics while suppressing noise. Lightweight Models: Recent advancements focus on lightweight models, such as LUNet and LMSC-DAE, which are optimized for real-time wearable applications. These models reduce computational complexity while maintaining effective noise suppression. For example: LUNet utilizes a U-shaped architecture to identify and reduce noise while preserving critical signal features (Reference [82]).

LMSC-DAE applies average pooling and point-wise convolution to achieve noise reduction with minimal resource requirements (Reference [80]).

Performance Highlights:

DAEs and GANs have shown superior performance in offline noise suppression, with SNR improvements ranging from 10 dB to 20 dB in various studies. Lightweight models like LUNet provide a balance between performance and computational efficiency, making them suitable for embedded systems.

2.4.2. Hybrid model methods

Researchers have investigated techniques for integrating deep learning methodologies with conventional algorithms. They tried combining sparse representation methods with deep neural networks with discrete wavelet transform and discrete cosine transform with noise-reducing self-coding networks [92]. These techniques preserved the signal's deep features [93] and achieved improved outcomes with reduced parameters and MAC values [94]. Utilizing a blend of conventional and contemporary methods improves the efficiency of the model and boosts its comprehensibility.

Unlike time domain, frequency domain, and time–frequency domain methods, which directly process signal characteristics, AI-based methods utilize advanced computational techniques such as deep learning to model noise characteristics and suppress them effectively. These methods often combine elements of traditional domains with novel algorithmic approaches, making them highly adaptable for complex and non-stationary noise suppression tasks. The distinction between signal domains (time, frequency, time–frequency) and methodological approaches (e.g., AI) is crucial for understanding the evolution of ECG signal processing. While AI-based methods do not represent a signal domain, their integration with traditional approaches enables robust noise suppression across multiple scenarios. This hybrid nature has positioned AI as a complementary and increasingly indispensable tool in ECG noise suppression. Hybrid models combine

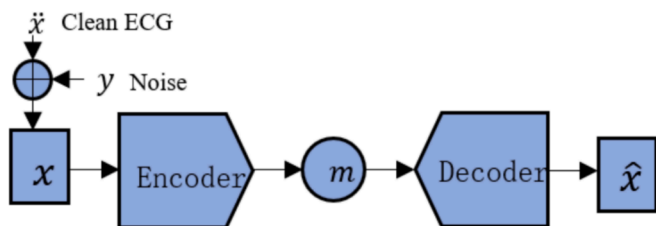


Fig. 7. Schematic diagram of noise-canceling self-encoder (. Adapted from Reference [69])

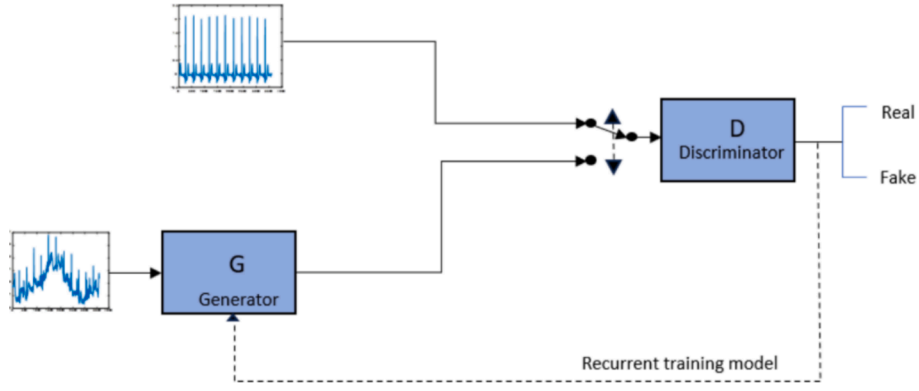


Fig. 8. Block diagram of adversarial network (. Adapted from Reference [83])

traditional signal processing techniques with deep learning approaches to leverage the strengths of both. Examples include:

Wavelet Transform + DAE: Combines the multi-resolution analysis capability of wavelets with the noise suppression power of DAEs, effectively reducing baseline wander and motion artifacts.

GAN + RNN: Integrates GAN's generative ability with RNN's temporal feature extraction, enhancing the robustness of noise suppression in dynamic environments.

Hybrid methods offer the following advantages:

Improved interpretability compared to purely data-driven models.

Enhanced adaptability to diverse noise conditions through the integration of domain knowledge. AI methods surpass traditional approaches in handling complex and composite noise. However, their higher computational costs and training requirements limit real-time applicability. Table 3 (cross-referenced here) summarizes the performance of various noise reduction techniques, highlighting the advantages of AI-based methods in terms of SNR improvement, noise-specific adaptability, and robustness against signal distortion.

2.5. Real-time processing capability

Real-time processing of dynamic ECG signals is the key to realizing real-time monitoring of wearable devices. Among the multiple algorithms reviewed in this paper, the following classes of methods are particularly suitable for real-time scenarios:

Adaptive filtering: such as Least Mean Square (LMS) and Recursive Least Squares (RLS) algorithms, these methods are able to adjust the filtering parameters in real time to adapt to signal changes in dynamic environments [13,15].

Wavelet transforms: both the discrete wavelet transform (DWT) and the smooth wavelet transform (SWT) can operate efficiently in real-time processing environments, and their multiscale decomposition properties make them suitable for dynamic signal processing [54,58].

Lightweight deep learning models: e.g., LUNet network architecture and Low Memory Connected Noise Reducing Self-Encoder (LMSC-DAE), which achieve high real-time performance by reducing the number of parameters and computational complexity [80,82]. Table 3 summarizes the real-time performance of different methods. Although the real-time performance of deep learning methods is slightly inferior to traditional methods, they have significant advantages in classification accuracy and noise reduction. Future research could explore more lightweight model design and hardware optimization.

3. Performance analysis

3.1. Database

The review research evaluated noise signals from the MIT-BIH Noise

Stress Test database, which included three common forms of noise: Baseline wander (bw), Muscle artifact (ma), and Electrode motion artifact (em). The noises were collected from half-hourly ambulatory ECG recordings obtained from individuals engaging in physical activity using a typical ECG recorder, leads, and electrodes [95]. ECG signal databases are crucial for noise suppression algorithms in ambulatory ECG signals. Table 2 outlines the top 10 frequently utilized open-source ECG signal databases.

3.2. Evaluation of indicators

The study categorizes evaluation criteria for algorithmic performance into two primary groups: statistically based metrics (1–4) and distance-based metrics (5–8) [3]. The signal-to-noise ratio and mean square error can obscure notable aberrations in the signal, leading to seemingly favorable outcomes [96]. Future studies should integrate these two types of measurements to evaluate algorithm performance more precisely. In the following equations, N represents the length of the ECG signal, x_i represents the clean ECG signal, x_n represents the noisy ECG signal, and x_m represents the noise-canceled ECG signal.

- 1) **Signal-to-noise ratio (SNR):** Signal-to-noise ratio refers to the difference between the signal-to-noise ratio after noise reduction and the signal-to-noise ratio of the original signal, and the larger the value of SNR_{imp} , the better the denoising effect.

$$SNR_{in} = 10 \times \log_{10} \left(\frac{\sum_{k=1}^N x_i^2}{\sum_{k=1}^N (x_n - x_i)^2} \right) \quad (1)$$

$$SNR_{out} = 10 \times \log_{10} \left(\frac{\sum_{k=1}^N x_i^2}{\sum_{k=1}^N (x_m - x_i)^2} \right) \quad (2)$$

$$SNR_{imp} = SNR_{out} - SNR_{in} \quad (3)$$

- 2) **Mean-Square Error (MSE):** Mean-Square Error is the mean-square difference between the noise-canceled signal and the original clean signal. The smaller the value of MSE, the better the denoising effect.

$$MSE = \frac{1}{N} \sum_{k=1}^N (x_i - x_m)^2 \quad (4)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{k=1}^N (x_i - x_m)^2} \quad (5)$$

Table 2
ECG database.

Database	Records	Length of record (minutes)	Leads	Sampling frequency (H z)	Source
MIT-BIH Arrhythmia Database ¹	47	30	2	360	Boston's Beth Israel Hospital
QT Database ²	105	15	2	250	Compiled from several existing databases
MIT-BIH Normal Sinus Rhythm Database ³	18	1440	2	128	Boston's Beth Israel Hospital
2018 China Physiological Signal Challenge ⁴	6877 train, 2954 test	0.25	12	500	11 hospitals
PhysioNet Computing in Cardiology Challenge 2017 ⁵	8528 train, 3658 test	0.5	1	300	AliveCor healthcare device
MIT-BIH ST Change Database ⁶	28	13–67	1–2	360	Boston's Beth Israel Hospital
PTB Diagnostic ECG Database ⁷	549	Several minutes	15	1000	National Metrology Institute of Germany
MIT-BIH Atrial Fibrillation Database ⁸	25	600	2	250	Boston's Beth Israel Hospital
St Petersburg INCART 12-lead Arrhythmia Database ⁹	75	30	12	257	11 hospitals
European ST-T ¹⁰	90	120	2	250	Thirteen research teams in eight countries

¹<https://physionet.org/content/mitdb/1.0.0/>.²<https://physionet.org/content/qtdb/1.0.0/>.³<https://physionet.org/content/nsrdb/1.0.0/>.⁴<https://2018.icbeb.org/Challenge.html>.⁵<https://www.physionet.org/content/challenge-2017/1.0.0/>.⁶<https://www.physionet.org/content/stdb/1.0.0/>.⁷<https://physionet.org/content/ptbdb/1.0.0/>.⁸<https://physionet.org/content/afdb/1.0.0/>.⁹<https://physionet.org/content/incartdb/1.0.0/>.¹⁰<https://www.physionet.org/content/edb/1.0.0/>.

- 3) Pearson correlation coefficient (PCC): Pearson correlation coefficient is used as a quantitative indicator to measure the linear relationship between two signals.

$$P = \frac{\text{cov}(x_i, x_m)}{\sigma x_i \bullet \sigma x_m} \quad (6)$$

- 4) Cross-Correlation Coefficient (C): Cross-Correlation Coefficient measures the strength of correlation between signals and is used for signal similarity testing.

$$C = \frac{\sum_{k=1}^N x_i x_m}{\sqrt{\sum_{k=1}^N x_i^2 \sum_{k=1}^N x_m^2}} \quad (7)$$

- 5) Maximum Absolute Deviation (MAD): Maximum Absolute Deviation measures the maximum absolute distance between the original signal and the filtered signal. the smaller the MAD value, the better the result.

$$MAD(x_i, x_m) = \max |x_i(n) - x_m(n)| \quad 1 \leq n \leq N \quad (8)$$

- 6) Sum of Squared Distances (SSD): Sum of Squared Distances calculates the sum of squared distances between the original signal and the filtered signal. the smaller the SSD value, the better the result.

$$SSD(x_i, x_m) = \sum_{n=1}^N (x_i(n) - x_m(n))^2 \quad 1 \leq n \leq N \quad (9)$$

- 7) Percentage Root-mean-square Difference (PRD): Percentage Root-mean-square Difference measures the degree of deviation of the noise-canceled signal relative to the original signal. The lower the value of PRD, the better the noise-canceling effect.

$$PRD(x_i, x_m) = \sqrt{\frac{\sum_{n=1}^N (x_m(n) - x_i(n))^2}{\sum_{n=1}^N (x_m(n) - \bar{x}_i)^2}} * 100\% \quad 1 \leq n \leq N \quad (10)$$

- 8) Cosine Similarity (CosSim): Cosine Similarity measures the similarity of two signals in terms of direction; the closer the cosine similarity is to 1, the more similar the two vectors are.

$$CosSim(x_i, x_m) = \frac{\sum_{n=1}^N x_i(n) x_m(n)}{\sqrt{\sum_{n=1}^N x_i^2(n)} \sqrt{\sum_{n=1}^N x_m^2(n)}} = \frac{\langle x_i(n), x_m(n) \rangle}{\|x_i\| \|x_m\|} \quad 1 \leq n \leq N \quad (11)$$

3.3. Performance analysis

Table 3 outlines the universality and specificity of dynamic ECG noise reduction methods designed for wearable devices. Adaptive filtering, noise detection, classic wavelet transform, and basic feature wave recognition algorithms are effective for real-time em noise reduction. The KF and SG filter are more effective in real-time noise reduction, with the Kalman filter being particularly notable in reducing ma noise. Traditional wavelet transform, SWT, and Eigenwave detection are effective methods for reducing bw noise in real time, particularly in the context of wearable devices for immediate analysis. Each method may offer some level of noise reduction for different types of noise. Artificial intelligence algorithms excel in noise reduction and successfully suppress complex noise. This approach is not designed for real-time applications and is better suited for offline analysis. The ECG signal was disrupted by noise and then examined quantitatively using Tables 4 to 11. PF or MCF is not needed to handle electromagnetic noise in conventional methods. The noise-reducing self-encoder, which combines average pooling, point-by-point convolution, and discrete wavelets, demonstrates optimal performance when deep learning methods' SNR falls within the −6 to 12 dB range. The signal numbered 118 in the MIT-BIH database underwent testing, and the noise reduction for ma noise described in the literature [46] was notably more substantial. The SG filter and the adversarial network are effective in suppressing AWGN noise when the SNR ranges from 0 to 10 dB. When dealing with low SNR or high levels of noise, the use of CEEMD in conjunction with Artificial Neural Networks or a diffusion model is recommended for reducing bw noise. The conditional adversarial network excels in reducing three distinct types of noise and the combined noise within the SNR range of 0 to 5 dB. A transformer equipped with an attention mechanism can be utilized to reduce composite noise. When taking into account the

Table 3
Summary of noise reduction algorithms.

Methods	references	Noise type	Advantage	disadvantage	topic-ality	Noise reduction effect
adaptive filtering	[15,16,17,18,19,20,21]	em	Real-time change adjustment parameters for non-static environments	Suitable reference signals are not easily accessible	√√	***
filters	[25,26,27,28]	em, bw	Capable of handling nonlinear and non-Gaussian problems with wide applicability	High computational costs and limited real-time applications	×	**
	[29,30]	ma	Lower computational requirements for real-time systems	For linear Gaussian models with limited ability to handle nonlinearities	√√√	***
	[31]	ma	Simple calculation, suitable for real-time data smoothing	Limited ability to handle complex dynamic noise	√√√	**
EMD	[33,34,35,36,37,38,39,41,42]	ma, bw	Highly adaptable	High computational requirements, not easy to realize real-time processing	×	***
noise detection	[43]	em	Directional processing of noise using domain knowledge and signal characterization	Complicated parameterization and adjustment	√√	***
	[44,45]	em	Combining multiple technologies to improve the overall treatment effect	High computational resource requirements and complexity	×	***
VMD	[47,48,49,50,51]	ma, bw	Signal features can be enhanced and signal patterns preserved	Higher complexity of implementation of optimization algorithms	×	**
WT	[54,55,56,57]	em, bw	Multi-scale analysis; efficient data compression	Edge effects, parameter selection	√√	***
	[58,59,60]	em, bw	Time scale information retention with high stability	poorly adaptive	√	***
	[62,63,64,65]	em, bw	Adaptive and effective signal decomposition	High computational requirements, not easy to implement in resource-limited devices	×	***
eigenwave detection	[71]	em, bw	Simple and effective, easy to implement and integrate	Not for patients with cardiac arrhythmias	√√√	***
	[72,73]	em	High noise resistance and low false detection rate	High computational requirements and real-time constraints	×	**
deep learning	[70,71,72,73,74,75,80,81,82,83,85,87,88,89]	em	High performance and adaptability, effectively retaining key signal features	Resource requirements and training costs are high, and the generalization ability of the model needs to be tested	×	****
	[74,82,83,85,87,89,93]	ma				
	[78,85]	AWGN				
hybrid model	[75,76,77,78,79,83,86,87,89,98]	em, bw, ma	Increased interpretability of medical applications; reduced model complexity and number of parameters	Fine tuning and adequate training are required to ensure that the advantages of both methods are fully utilized	√	****
	[74,82, 83, 85, 87, 88 89,90,91,93]	bw				
	[92,93,94]	em, bw, ma				

Note: √: Higher number means better real-time *: Higher number means better noise suppression × : Indicates poor real-time.

performance and power consumption of algorithm, lightweight models are a good option.

The detailed analyses of Tables 4–11 reveal significant advancements in ECG noise suppression, particularly in the integration of traditional, knowledge-based, and deep learning methods. The trajectory of progress indicates a shift toward hybrid models that balance accuracy, efficiency, and adaptability. Future research should focus on addressing the limitations of existing datasets, improving model interpretability, and optimizing methods for real-time wearable applications. Table 4 compares the performance of traditional filtering methods for ECG noise suppression. While adaptive filters such as LMS and RLS demonstrate significant effectiveness for baseline wander and industrial frequency interference, their performance declines for non-stationary noise, such as motion artifacts. Additionally, parameter tuning remains a critical challenge, requiring expert knowledge for optimal performance. Recent advancements have focused on hybrid methods that integrate traditional filtering with machine learning to improve adaptability and robustness, marking a promising trajectory for future research. Table 5 highlights the emerging trend of knowledge-based

filtering methods. These approaches leverage domain-specific knowledge, such as ECG signal morphology and physiological patterns, to design robust filters. However, their reliance on predefined rules limits their flexibility in handling unanticipated noise types. Current research is exploring the integration of knowledge-based systems with AI techniques to create hybrid models capable of both rule-based precision and data-driven adaptability.

Tables 6–9 summarize the advancements in deep learning models for ECG noise suppression, covering CNNs, RNNs, GANs, and autoencoders. While CNNs excel in spatial feature extraction and offer high accuracy, their computational demands make them less suitable for real-time wearable applications. RNNs, on the other hand, effectively capture temporal dependencies but face challenges with long-term dependencies and training stability. GANs have shown promising results in generating clean ECG signals but require large training datasets and are computationally intensive. Autoencoders offer a balance between efficiency and performance, making them ideal for embedded systems. The trajectory of progress suggests a growing focus on lightweight, interpretable models optimized for real-time applications. Table 10 presents a

Table 4
Electrode motion artifact suppression methods(MIT-BIH Database).

Recorded Signals	Average (MIT-BIH)								
SNR_{in}	−5dB			5 dB			0 dB		
Noise reduction methods	SNR_{imp}	MSE	C	SNR_{imp}	MSE	C	SNR_{imp}	MSE	C
Multi denoise[43]	11.43	0.025	0.87	9.14	0.004	0.98	—	—	—
IEMD-ATD[38]	2.79	—	0.59	1.69	—	0.89	—	—	—
Particle Filter[25]	21.12	—	0.995	11.94	—	0.965	—	—	—
Monte Carlo Filter[28]	—	—	—	15.01	0.003	0.981	—	—	—
EWT-WT[62]	—	—	—	—	—	—	9.74	0.0018	—
SNR_{in}	−6 dB			6 dB			12 dB		
Noise reduction methods	SNR_{imp}	C	PRD, %	SNR_{imp}	C	PRD, %	SNR_{imp}	C	PRD, %
RNN −DNN[71]	—	—	—	22.48	0.626	—	18.20	0.80	—
LMSC-DAE[80]	19.30	—	86.04	12.74	—	46.89	8.78	—	37.18
FCN[75]	18.22	—	97.67	11.95	—	52.39	7.75	—	42.07
CNN-LSTM[97]	18.71	—	92.90	8.94	—	71.01	2.83	—	71.71
CPDAE[81]	23.68	—	51.20	24.99	—	12.81	21.38	—	9.28
SNR_{in}	0 dB			1.25 dB			5 dB		
Noise reduction methods	SNR	RMSE		SNR	RMSE		SNR	RMSE	
SCED-Net[74]	26.04	0.025		27.15	0.022		28.44	0.021	
LUNet[82]	24.38	0.018		24.35	0.018		24.39	0.017	
DAE[88]	21.56	0.037		21.96	0.035		22.78	0.031	
ADnCNN[89]	32.94	0.007		35.25	0.005		36.00	0.005	
AN[83]	32.38	0.0102		32.90	0.0096		33.79	0.009	
GAN[85]	—	—		—	—		26.97	0.027	
CGAN[87]	40.09	0.002		43.28	0.002		41.75	0.003	
DW-CNN[93]	7.06	0.311		7.54	0.307		8.06	0.304	

Table 5
Electrode motion artifact suppression methods(MIT-BIH NSR Database).

Recorded Signals	Average (MIT-BIH NSR Database)								
Noise reduction methods	SNR_{imp}	C	PRD	SNR_{imp}	C	PRD	SNR_{imp}	C	PRD
DWT-DAE[94]	25.29	—	72.53	18.80	—	42.16	13.90	—	36.53
DCT-DAE[94]	25.13	—	72.71	19.24	—	41.45	14.72	—	34.51

Table 6
Gaussian white noise suppression methods(MIT-BIH Database).

Recorded Signals	Average (MIT-BIH)								
SNR_{in}	0 dB			5 dB			10 dB		
Noise reduction methods	SNR_{imp}	MSE	PRD	SNR_{imp}	MSE	PRD	SNR_{imp}	MSE	PRD
LDASG[31]	15	0.022	17	20	0.01	10	24	0.002	7
MEMD-NLM[37]	11.70	—	13	9.50	—	9.80	7.95	—	6
VMD-NLM-WT[47]	11.80	0.002	11.83	10.60	0.001	7.69	9.16	0.0003	5.14
TWO-VMD[50]	—	—	—	6.01	0.004	28.50	5.81	0.001	16.30
ACDAE[78]	14.63	0.0005	9.8	—	—	—	—	—	—
IEMD-ATD[38]	—	—	—	5.78	—	—	3.19	—	—
EEMD-LM[39]	—	—	—	—	—	—	8.45	—	—
Recorded Signals	100								
EMD-NLM[37]	11.7	—	13	9	—	9	7.8	0.0005	6
EMD-SE[33]	—	—	—	—	—	—	7.75	0.018	—
EEMD-LM[39]	—	—	—	—	—	—	11.78	—	—
Recorded Signals	107								
ECG-GAN[85]	—	—	—	14.82	0.0023	11.43	16.30	0.0016	9.63
EMS-SE[33]	—	—	—	—	—	—	7.98	—	—

comparative analysis of publicly available ECG datasets. While datasets such as MIT-BIH and PhysioNet remain widely used, their limited annotations for noise types constrain the development of noise-specific algorithms. Recent efforts have focused on generating synthetic datasets or augmenting existing datasets with labeled noise segments, paving the way for more targeted algorithm development. Table 11 provides a holistic view of the performance of various noise suppression methods across different metrics. While traditional and knowledge-based methods maintain efficiency and interpretability, deep learning

models excel in handling complex and composite noise. However, the trade-offs between accuracy, computational cost, and generalizability remain a key area for future research. This highlights the need for hybrid approaches that leverage the strengths of different methodologies.

Table 7
Muscle artifact suppression methods(MIT-BIH Database).

Recorded Signals	Average (MIT-BIH)					
	0 dB		1.25 dB		5 dB	
SNR_{in}						
Noise reduction methods	SNR	RMSE	SNR	RMSE	SNR	RMSE
SCED-Net[74]	25.23	0.024	25.95	0.023	27.24	0.021
LUNet[82]	23.73	0.019	23.00	0.022	24.33	0.018
ADnCNN[89]	35.36	0.005	35.87	0.005	38.03	0.004
AN [83]	35.14	0.0088	35.87	0.0081	37.23	0.007
GAN[85]	—	—	—	—	29.32	0.020
CGAN[87]	39.09	0.0033	42.14	0.0022	40.59	0.003
DW-CNN[93]	7.74	0.292	7.70	0.3	8.04	0.285
Recorded Signals	118					
Noise reduction methods	SNR_{imp}		MSE			
EEMD-LM[39]	14.53		2.6392			
VMD-NLM-WT[47]	14.71		2.2599			
GOA-VMD-DWT[49]	16.22		2.2557			

Table 8
Baseline wander suppression methods(MIT-BIH Database).

Recorded Signals	Average (MIT-BIH)					
	0 dB		1.25 dB		5 dB	
SNR_{in}						
Noise reduction methods	SNR	RMSE	SNR	RMSE	SNR	RMSE
SCED-Net[74]	27.87	0.023	28.66	0.023	29.96	0.022
LUNet[82]	25.06	0.016	26.07	0.015	26.53	0.013
ADnCNN[89]	30.60	0.007	35.64	0.005	36.92	0.004
AN [83]	34.83	0.006	35.31	0.006	36.29	0.005
GAN[85]	—	—	—	—	29.70	0.023
CGAN[87]	39.59	0.003	42.12	0.002	41.10	0.003
DW-CNN[93]	7.71	0.294	8.14	0.299	7.69	0.275
Recorded Signals	103					
Noise reduction methods	SNR_{imp}	C	SNR_{imp}	C	SNR_{imp}	C
EWT[63]	16.81	0.9672	12.90	0.9749	8.30	0.9773
EMD-LWT[34]	14.43	0.9411	14.41	0.9815	11.05	0.9860
Noise reduction methods	SNR_{imp}		MSE			
CEEMDAN-WT [41]	15.37		0.0011			
Auto-encoder[88]	18.89		0.0006			
ECG-GAN[85]	25.70		0.0004			
VMD-SSA-SVD[51]	18.48		0.043			

4. Discussion

4.1. ECG data sets

Various investigations have utilized varying quantities of leads, acquisition periods, sampling rates, and data sources, as indicated in Table 2. The absence of standardization hinders the comparison of outcomes across different data sets. Due to the difficulty in acquiring high-quality data, such as long-term ECG recordings and their annotations, many current studies continue to depend on the MIT-BIH

arrhythmia database, which was compiled four decades ago. MIT-BIH arrhythmia database exhibit several limitations that hinder the development of robust algorithms for wearable devices:

Limited Noise Diversity: Most datasets are collected in controlled environments, lacking the complexity of real-world noise scenarios such as motion artifacts during physical activities.

Inconsistent Annotations: Annotation standards vary across datasets, making it difficult to benchmark algorithms consistently. For example, noise labels are often incomplete or imprecise, limiting their utility for supervised learning.

Underrepresentation of Demographic and Device Variability: Many datasets fail to capture diverse populations and different device configurations, reducing the generalizability of trained models.

Recommendations for Improvement: Enhanced Annotation Standards: Establishing unified guidelines for noise categorization and labeling will improve the comparability and utility of ECG datasets.

Inclusion of Real-World Scenarios: Expanding datasets to include recordings from ambulatory monitoring under various conditions, such as exercise or daily activities, will better represent wearable device use cases.

Collaborative Dataset Creation: Encouraging collaborations among academia, industry, and clinical institutions can lead to the creation of large-scale, publicly accessible datasets with diverse demographics and noise conditions. Future efforts should prioritize the creation of high-quality long-term ECG datasets to support new and innovative research.

Although Table 2 lists databases primarily containing 12-lead or 6-lead ECG signals, many of the algorithms reviewed in this work are inherently adaptable to 1-lead or 3-lead configurations commonly used in wearable ECG devices. For instance, adaptive filtering methods such as LMS and RLS, which rely on reference signal correlations rather than the number of leads, have demonstrated effective noise suppression in low-lead systems. Additionally, lightweight models like LMSC-DAE and LUNet have been specifically designed with computational efficiency in mind, making them suitable for single-lead or three-lead setups where processing power and data availability are limited. Several reviewed studies have also validated their algorithms on datasets or scenarios involving 1-lead or 3-lead ECG signals. For example, algorithms designed for ambulatory monitoring often focus on single-lead signals due to the constraints of wearable device configurations. This indicates the generalizability of the noise suppression methods analyzed in this review to wearable devices employing fewer leads.

4.2. Analogical learning

Analogical learning is a computational paradigm inspired by human reasoning, which draws on similarities between different domains or tasks to inform decision-making. In the context of ECG signal processing for wearable devices, analogical learning can be applied to transfer knowledge from well-annotated datasets (e.g., clinical-grade ECG recordings) to less structured, real-world wearable ECG data. Due to the similarity between motion artifact in EEG and ECG signals in terms of frequency overlap and their significantly greater amplitude compared to clear EEG signals, ECG signal processing methods can be utilized for category learning. Hence, they can be mitigated by channel artifact-based reference regression and spatial filtering methods [99], inertial sensing techniques [100], and Conventional Correlation Analysis (CCA) approaches [101]. In [102], Shukla introduced a new approach that

Table 9
Baseline wander suppression methods (QT Database).

Recorded Signals	Average(QT Database)							
	0.2 to 0.6				1.0 to 1.5			
noise								
Noise reduction methods	SSD	MAD	PRD	CosSim	SSD	MAD	PRD	CosSim
LANLD[90]	2.4	0.29	34.03	0.948	5.7	0.40	53.78	0.888
DeScoD-ECG[91]	2.0	0.21	27.48	0.962	4.1	0.36	43.32	0.920

Table 10

Composite noise (bw, em, ma) suppression methods(MIT-BIH Database).

Recorded Signals	Average (MIT-BIH)								
SNR_{in}	0 dB			1.25 dB			5 dB		
Noise reduction methods	SNR_{imp}	RMSE	PRD	SNR_{imp}	RMSE	PRD	SNR_{imp}	RMSE	PRD
ADnCNN[89]	32	0.0094	6.94	32.25	0.0083	6.07	30.60	0.0068	4.98
GAN[83]	30.50	0.0137	—	29.70	0.0131	—	26.71	0.0122	—
CGAN[87]	39.49	0.0031	—	—	—	—	36.28	0.0025	—
SNR_{in}	−1dB			7 dB					
Noise reduction methods	SNR_{imp}	RMSE	PRD	SNR_{imp}	RMSE	PRD	SNR_{imp}	RMSE	PRD
FCN-DAE[75]	15.49	0.063	19.68	10.54	0.056	14.05			
GAN-LSTM[86]	19.25	0.035	12.91	13.11	0.028	10.35			

Table 11

Composite noise (bw, em, ma) suppression methods(QT Database).

Recorded Signals	Average(QT Database)							
noise	0.2 to 0.6				1.0 to 1.5			
Noise reduction methods	SSD	MAD	PRD	CS	SSD	MAD	PRD	CS
CDAE-BAM[77]	9.44	0.758	112.17	0.975	13.71	0.754	109.94	0.970
SNR_{in}	−6–0 dB				6–12 dB			
Noise reduction methods	SNR_{imp}	RMSE	CosSim		SNR_{imp}	RMSE	CosSim	
DRDNN[98]	8.26	0.14	0.85		12.26	0.07	0.93	
ACDAE[78]	11.48	0.11	0.90		16.41	0.05	0.98	
TCDAE[79]	12.08	0.07	0.95		14.43	0.06	0.98	
CDAE-BAM[77]	11.42	0.11	0.90		16.17	0.05	0.98	

combines Gaussian Elimination CCA (GECCA) with EEMD using a modified cascade strategy. This method includes incorporating a unique median filter before the GECCA phase to enhance the effectiveness of artifact removal. While methods like EEMD, CEEMD, and CEEMDAN have advanced in addressing the modal aliasing issue, they lack the ability to accurately identify extremal points and envelopes. To address these problems, Ranjan et al. [103] introduced an enhanced adaptive EMD approach to regulate the ideal number of repetitions through soft-thresholding adaptive selection of the stopping criterion. Despite its potential, the integration of analogical learning in wearable ECG systems remains underexplored. Future research should focus on developing algorithms that can effectively harness analogical reasoning to enhance model robustness and adaptability under diverse conditions.

4.3. Interpretability

Interpretability is a critical aspect of wearable ECG systems, as it bridges the gap between algorithmic decisions and clinical insights. Recent advancements have focused on developing interpretable models that allow clinicians and end-users to understand the reasoning behind noise suppression and signal classification outputs. Deep learning models have shown substantial advancements in signal processing and have displayed exceptional generalization capabilities in the past several years. Its interpretability is significantly lacking compared to traditional approaches, making it challenging for medical experts to accept diagnostic data without interpretation, thereby restricting its use in the medical field. If the physiological reasons behind the predictions produced by supervised AI algorithms can be comprehended, the judgments are likely to be clinically acceptable [104]. Research has been concentrated on enhancing the comprehensibility of deep learning models for analyzing ECG signals. Researchers have developed globally interpretable predictors to overcome the issue of a high number of model parameters and to simulate the behavior of “black box” models [105]. This involves incorporating rule learning and introducing Shap ways to measure the interpretation of each feature [106] or putting expert features into the network to enhance the interpretability of the

model. In addition, the waveforms were visualized using the class activation gradient mapping technique [107] to make them logically consistent with the physician’s judgment. Complex models’ predictions in research can be elucidated by utilizing a simpler model or by directly creating an interpretable model, such as by integrating traditional techniques or including attentional mechanisms in the network. By advancing interpretability in wearable ECG systems, future solutions can improve user trust, support clinical decision-making, and enable widespread adoption in health monitoring applications.

4.4. Future developments

Noise suppression of ECG signals under motion is the basis for subsequent studies such as feature extraction and waveform detection. Algorithms in AI, intense learning and hybrid models demonstrate superior noise reduction compared to time, frequency, and time–frequency domains. If the real-time capability of the models can be enhanced, these algorithms could be integrated into wearable devices. To overcome battery capacity restrictions, models must utilize lossless data compression for efficient high-speed transmission and long-term data storage. Unfortunately, only a few algorithms with compression capabilities are currently available. Future studies should investigate model compression methods such as knowledge distillation, weight sharing, and exact encoding of network weights to reduce model size. Migrating learning can effectively lower the expenses associated with model training and deployment, facilitating the development of cost-efficient, time-saving, and pragmatic algorithms. Challenges in implementing wearable technology in EMS practice include issues related to data accuracy, reliability, transmission speed, sharing capabilities, patient adherence to using wearable devices, and product interoperability. With the continuous development of sensing devices and information technology, wearable devices are gradually acquiring more complex and diversified functions and will become an integral part of prehospital emergency care. These devices enhance real-time monitoring and early warning capabilities and facilitate the widespread use of portable intelligent ECG monitoring devices.

While this review has demonstrated the generalizability of noise suppression algorithms to wearable ECG systems with fewer leads, further research is needed to specifically tailor these methods for 1-lead and 3-lead configurations. These configurations are more susceptible to motion artifacts and electrode displacement noise due to their minimalist design. Future work could focus on:

1. Developing noise suppression algorithms that explicitly address the unique challenges of single-lead and three-lead ECG signals, such as motion artifacts and baseline drift.
2. Optimizing AI-based models for these configurations by leveraging domain-specific knowledge, such as the spatial characteristics of low-lead signals.
3. Conducting experimental validations on datasets specifically collected from 1-lead and 3-lead wearable ECG devices to further enhance the applicability of the reviewed algorithms.

4.5. Multi-Functional algorithms for ECG signal processing

In recent years, researchers have gradually discovered that some algorithms can be used not only for noise reduction of dynamic ECG signals, but also for data compression or other operations at the same time. The application of such multifunctional algorithms not only improves the practicality of the algorithms, but also reduces the computational and storage resource requirements of wearable devices. The following are several representative multifunctional algorithms and their applications in ECG signal processing:

Wavelet transform is a typical time–frequency domain analysis tool, which can not only extract the main features in ECG signals for noise reduction, but also effectively compress the signals through multi-scale decomposition techniques. Literature [54] and [62] demonstrated the effect of wavelet transform in noise reduction and data compression of dynamic ECG signals, respectively, and proved its important application value in multifunctional signal processing. However, the edge effect of the wavelet transform and its dependence on the selection of wavelet basis functions limit its further generalization.

Sparse representation algorithms can not only effectively remove the noise in ECG signals, but also significantly reduce the amount of data storage and transmission by finding the sparse characteristics of signals. For example, in dynamic ECG signal processing scenarios, sparse representation can efficiently encode signals for noise reduction and compressed sensing to support real-time ECG monitoring [92]. This approach enables efficient data processing while maintaining the signal structure.

VMD, as a non-recursive signal decomposition technique, is capable of adaptively separating the individual mode components in a signal and is used to eliminate noise at different frequencies. In addition, VMD enables efficient compression by preserving the main feature components of the ECG signal [47]. VMDs that incorporate optimization algorithms such as particle swarm optimization or grasshopper optimization algorithms further improve their versatile performance [48].

Deep learning models also show the potential to achieve both noise reduction and compression. For example, noise-reducing self-encoders (DAEs) and generative adversarial networks (GANs) can drastically reduce signal storage space requirements while maintaining the main features of the signal [80,85]. In addition, hybrid models (e.g., models combining wavelet transform and deep learning) have achieved excellent performance in multifunctional signal processing [94].

The application of multifunctional algorithms in dynamic ECG signal processing not only significantly improves the noise reduction and compression efficiency, but also provides more possibilities for future wearable devices. However, these algorithms still face many challenges in terms of real-time performance, computational complexity, and hardware resource limitations. For example, deep learning models usually require large amounts of computational resources and training

data, which may not be suitable for low-power wearable devices. Future research should focus on the development of lightweight models and the optimization of multifunctional algorithms for adaptation in real-world scenarios. By exploring multifunctional algorithms, we can process dynamic ECG signals more efficiently while reducing the storage and energy requirements of hardware devices, paving the way for the wide application of wearable devices.

4.6. Significance and development of real-time capability

The ability to process ambulatory ECG signals in real-time is fundamental to the clinical application of wearable devices. This paper reviews the applicability of a variety of methods in a real-time environment, with particular emphasis on the following points: Real-time advantages of traditional algorithms: methods such as adaptive filtering and wavelet transform have low computational complexity and can be directly embedded in wearable devices for real-time signal noise reduction. The potential of lightweight deep learning: with the development of model compression techniques (e.g., knowledge distillation and pruning techniques), lightweight deep learning models are gradually becoming a viable solution for real-time applications. Relationship between real-time processing and classification: real-time noise reduction in the signal preprocessing stage is crucial for the accuracy and robustness of classification frameworks. Future research should explore end-to-end real-time signal processing and classification systems to further improve performance in dynamic environments. Real-time computational capability is critical for wearable ECG devices to ensure continuous monitoring and timely response to cardiac events. In this review, several algorithms are identified and evaluated for their suitability in real-time processing environments. Adaptive filtering techniques, such as LMS and RLS, dynamically adjust filtering parameters, making them well-suited for processing dynamic signals with minimal delay. Lightweight models, including LMSC-DAE [80] and LUNet [82], leverage architectural optimizations to reduce computational overhead and memory usage, enabling their deployment on resource-constrained wearable devices. These methods, while differing in complexity, share a focus on balancing noise suppression performance with real-time feasibility. For example, LMSC-DAE achieves real-time capability through parameter reduction, while adaptive filtering methods utilize low-complexity iterative updates to handle non-stationary noise effectively. The insights gained from this literature survey provide a foundation for developing algorithms tailored to wearable ECG systems, ensuring they meet the stringent demands of real-time operation.

Wearable ECG devices predominantly utilize 1-lead or 3-lead configurations due to their compact design and low power consumption. Although the databases reviewed in this study are primarily based on 12-lead or 6-lead recordings, the algorithms discussed are largely independent of the number of leads. For instance, the frequency-domain and time–frequency-domain approaches, such as wavelet transforms, primarily process the signal's temporal and spectral characteristics, which remain consistent across different lead configurations. Similarly, adaptive filtering techniques and AI-based models focus on signal features rather than spatial lead information, allowing them to be effectively applied to single-lead and three-lead ECG signals. Furthermore, the lightweight architectures of models like LMSC-DAE and LUNet ensure compatibility with the limited computational resources of wearable devices. These models can process real-time signals in low-lead systems while maintaining high noise suppression performance. Future research could explore further optimization of these algorithms for 1-lead or 3-lead settings, with a focus on reducing the computational and memory requirements specific to wearable ECG systems.

4.7. Comparative evaluation of noise suppression methods

To provide a comprehensive understanding of the noise suppression methods discussed, this section evaluates the strengths and weaknesses

of methods from different domains.

Time Domain Methods.

Strengths:

- Low computational complexity, making them suitable for real-time applications in wearable devices.
- Effective for reducing low-frequency noises like baseline wander.

Weaknesses:

- Limited performance for high-frequency noise (e.g., muscle artifacts).
- Dependence on accurate parameter tuning (e.g., adaptive filter weights) to ensure effectiveness.

Frequency Domain Methods.

Strengths:

- Robust against specific noise types such as industrial frequency interference.
- Provide high precision in isolating noise within a predefined frequency range.

Weaknesses:

- Inflexible for non-stationary noise due to fixed frequency assumptions.
- Computationally intensive for resource-constrained wearable devices.

Time-Frequency Domain Methods.

Strengths:

- Capable of handling non-stationary noise through multi-resolution analysis (e.g., wavelet transform).
- Preserves both time-domain and frequency-domain characteristics of ECG signals.

Weaknesses:

- Susceptible to edge effects and parameter selection (e.g., wavelet basis).
- Higher computational complexity than traditional time-domain methods.

Artificial Intelligence Methods.

Strengths:

- Superior noise suppression performance, especially for complex and composite noise.
- Capable of learning signal patterns adaptively, enhancing generalization.

Weaknesses:

- High computational cost and memory requirements, limiting real-time applicability.
- Limited interpretability of deep learning models, which may hinder clinical adoption.

4.8. Power constraints and energy efficiency in noise suppression algorithms

1. **Trade-offs Between Power Consumption and Algorithm Complexity:** Wearable devices are inherently limited by their energy supply, which poses significant challenges for real-time algorithms

with high computational complexity. Adaptive filtering methods, such as LMS and RLS, offer dynamic weight updates that enable effective noise suppression but require substantial computational resources. Similarly, deep learning-based methods, such as GANs and DAEs, excel in noise suppression but demand high energy due to the multilayered neural network operations, making them less suitable for energy-constrained devices.

2. **Energy Efficiency of Lightweight Algorithms:** To address these challenges, lightweight algorithms have been developed to optimize energy efficiency while maintaining acceptable performance:

LMSC-DAE: This model achieves energy efficiency by reducing parameter count and utilizing efficient point-wise convolutions. Its architecture minimizes memory usage and computational load, making it suitable for wearable devices.

LUNet: Designed with a U-shaped architecture, LUNet identifies and processes noise-specific regions, thereby avoiding unnecessary computations and enhancing energy efficiency.

3. Hardware and Algorithm Co-Optimization:

Energy-efficient hardware design plays a crucial role in overcoming power constraints:

Model Quantization: Reduces floating-point computations by quantizing weights and activations, significantly lowering power consumption.

Hardware Acceleration: Implements algorithms on low-power FPGA or ASIC platforms, providing dedicated hardware support for energy-intensive operations.

4. Future Research Directions

Hybrid Models for Energy Efficiency: Combining traditional low-power methods (e.g., time-frequency filtering) with deep learning models to balance performance and power consumption.

Data Compression and Transmission Optimization: Leveraging compression sensing and sparse coding to reduce energy demands during data transfer and storage.

Potential of Transfer Learning: Reducing training costs and improving model efficiency in resource-constrained environments by applying transfer learning techniques.

5. Conclusion

This review systematically analyzes the noise suppression methods for dynamic ECG signals, presents a detailed evaluation of the applicability and performance of different algorithms, and the main conclusions are as follows:

Diversity of noise suppression methods for dynamic ECG signals: This paper summarizes the noise reduction methods in the time domain, frequency domain, time-frequency domain, and artificial intelligence domain, and finds by comparative analysis:

Traditional methods (e.g. adaptive filtering, wavelet transform) have obvious advantages in real-time and computational efficiency, and are suitable for embedded wearable devices. Artificial intelligence domain methods (e.g., noise-reducing self-encoder, generative adversarial network) have excellent performance in noise reduction, but have higher real-time and computational resource requirements and are more suitable for offline analysis. Correlation between real-time capability and classification performance: In this paper, we pay special attention to algorithms with high real-time processing capability, and find that efficient signal preprocessing plays a significant role in improving classification performance. For example, adaptive filtering combined with lightweight deep learning models can achieve real-time noise reduction and classification in dynamic environments. Limitations of Existing Research: Current dynamic ECG signal processing faces the

following challenges: The lack of uniform standards for datasets limits the accuracy of cross-study comparisons. The design of lightweight models still needs to weigh the noise reduction effect against the computational cost. The lack of interpretability of deep learning models limits their clinical applications.

Future Research Directions: Dataset development: it is recommended to develop high-quality and richly labeled long-duration ambulatory ECG datasets to enhance the generalizability and innovation of the study.

Lightweight model optimization: exploring model compression techniques (e.g., knowledge distillation and pruning) to enhance the real-time and embedded adaptability of deep learning models.

Interpretability enhancement: Combining traditional methods and deep learning techniques to improve the medical interpretability of models for clinical applications.

Real-Time Performance Optimization: Many existing methods, particularly deep learning approaches, face computational challenges when implemented in wearable devices. Future work should focus on lightweight model architectures and hardware acceleration to enable real-time applications.

Standardization of ECG Datasets: A unified framework for dataset annotation, noise categorization, and benchmarking would greatly enhance the comparability and reproducibility of noise suppression methods. Collaborative efforts among academia, industry, and clinical institutions will be essential in achieving this goal.

Multi-functional algorithm development: design multi-functional algorithms that can simultaneously realize noise reduction, classification and compression to further reduce device power consumption and storage requirements.

In summary, this paper provides a comprehensive technology assessment and direction guide for the future development of dynamic ECG signal processing. The potential of wearable devices for ambulatory ECG monitoring will be further unlocked by driving advances in real-time, interpretability and efficiency.

CRediT authorship contribution statement

Juya Zhang: Writing – original draft, Project administration, Investigation, Formal analysis. **Yu Guo:** Visualization, Resources. **Xinming Dong:** Supervision. **Tong Wang:** Validation. **Jinhai Wang:** Methodology, Conceptualization. **Xin Ma:** Funding acquisition. **Huiquan Wang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This work was supported by the National Key Research and Development Program of China (Grant No. 2023YFC3011802).

Data availability

No data was used for the research described in the article.

References

- [1] G.A. Roth, C. Johnson, A. Abajobir, et al., Global, regional, and national burden of cardiovascular diseases for 10 causes, 1990 to 2015 [J], *J. Am. Coll. Cardiol.* 70 (1) (2017) 1–25.
- [2] S.S. Martin, A.W. Aday, Z.I. Almarzooq, et al., 2024 heart disease and stroke statistics: a report of US and global data from the american heart association [J], *Circulation* 149 (8) (2024) e347–e913.
- [3] ROMERO F P, ROMAGUERA L V, VÁZQUEZ-SEIDEDOS C R, et al. Baseline wander removal methods for ECG signals: A comparative study [J]. *arXiv preprint arXiv:1807.11359*, 2018.
- [4] M.A. Serhani, H. T E K, H. Ismail, et al., ECG monitoring systems: review, architecture processes, and key challenges [J], *Sensors (Basel)* 20 (6) (2020).
- [5] M. GUALSAQUÍ, I. VIZCAÍNO, V.G.P. ROSERO, et al., ECG signal denoising using discrete wavelet transform: a comparative analysis of threshold values and functions [J], *Maskana* 9 (1) (2018) 105–114.
- [6] ZHANG H, ZHANG S, JIN Q, et al. Motion artifact suppression in ambulatory ECG with feed forward combined adaptive filter; proceedings of the 2016 Computing in Cardiology Conference (CinC), F, 2016 [C]. IEEE.
- [7] H.N. Ingabire, K. Wu, J.T. Amos, et al., Analysis of ECG signals by dynamic mode decomposition [J], *IEEE J. Biomed. Health Inform.* 26 (5) (2021) 2124–2135.
- [8] DHAWAN A. Effective Delineation of ECG Signals for Operative Neural Disease Recognition and Classification; proceedings of the 2023 1st International Conference on Innovations in High Speed Communication and Signal Processing (IHCSPP), F, 2023 [C]. IEEE.
- [9] U. Satija, B. Ramkumar, M.S. Manikandan, A review of signal processing techniques for electrocardiogram signal quality assessment [J], *IEEE Rev. Biomed. Eng.* 11 (2018) 36–52.
- [10] I. Dotsinsky, An approach to successful power-line interference suppression in ECG signals [J], *Int. J. Bioautomation* 26 (1) (2022) 83.
- [11] L. FRÖLICH, I. DOWDING, Removal of muscular artifacts in EEG signals: a comparison of linear decomposition methods [J], *Brain Inf.* 5 (1) (2018) 13–22.
- [12] MAKDESSY C, CAO H, PEYRODIE L, et al. A comparative analysis of ECG denoising methods; proceedings of the 2020 IEEE 20th International Conference on Bioinformatics and Bioengineering (BIBE), F, 2020 [C]. IEEE.
- [13] M.M.U. Faiz, I. Kale, Removal of multiple artifacts from ECG signal using cascaded multistage adaptive noise cancellers [J], *Array* 14 (2022) 100133.
- [14] H.Y. Mir, O. Singh, ECG denoising and feature extraction techniques—a review [J], *J. Med. Eng. Technol.* 45 (8) (2021) 672–684.
- [15] S. Yu, Q. Song, S. Liu, Motion artifact cancellation from a single channel SCG using adaptive forgetting factor recursive least square filter [J], *IEEE Access* (2020).
- [16] CASTAÑO USUGA F A, GISSEL C, HERNÁNDEZ A M. 2022 Motion Artifact Reduction in Electrocardiogram Signals Through a Redundant Denoising Independent Component Analysis Method for Wearable Health Care Monitoring Systems: Algorithm Development and Validation [J]. *JMIR Medical Informatics*. 10(11): e40826.
- [17] X. An, G.K. Stylios, Comparison of motion artefact reduction methods and the implementation of adaptive motion artefact reduction in wearable electrocardiogram monitoring [J], *Sensors* 20 (5) (2020) 1468.
- [18] F. Xiong, D. Chen, Z. Chen, et al., Cancellation of motion artifacts in ambulatory ECG signals using TD-LMS adaptive filtering techniques [J], *J. Vis. Commun. Image Represent.* 58 (2019) 606–618.
- [19] S.-Y. Lee, P.-H. Su, Y.-W. Hung, et al., Motion artifact reduction algorithm for wearable electrocardiogram monitoring systems [J], *IEEE Trans. Consum. Electron.* (2023).
- [20] GHALEB F A, KAMAT M, SALLEH M, et al. Motion Artifact Reduction Algorithm Using Sequential Adaptive Noise Filters and Estimation Methods for Mobile ECG; proceedings of the Recent Trends in Information and Communication Technology: Proceedings of the 2nd International Conference of Reliable Information and Communication Technology (IRICT 2017), F, 2018 [C]. Springer.
- [21] XU Z, WANG Y, TIAN X, et al. Motion artifact removal based on stationary wavelet transform and adaptive filtering for wearable ECG monitoring; proceedings of the The proceedings of the 16th Annual Conference of China Electrotechnical Society: Volume I, F, 2022 [C]. Springer.
- [22] B. Pholpoke, T. Songthawornpong, W. Wattapanitch, A micropower motion artifact estimator for input dynamic range reduction in wearable ECG acquisition systems [J], *IEEE Trans. Biomed. Circuits Syst.* 13 (5) (2019) 1021–1035.
- [23] R. Sameni, M.B. Shamsollahi, C. Jutten, et al., A nonlinear Bayesian filtering framework for ECG denoising [J], *IEEE Trans. Biomed. Eng.* 54 (12) (2007) 2172–2185.
- [24] NATHAN V, AKKAYA I, JAFARI R. A particle filter framework for the estimation of heart rate from ECG signals corrupted by motion artifacts; proceedings of the 2015 37th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), F, 2015 [C]. IEEE.
- [25] S. Banerjee, G.K. Singh, A new moving horizon estimation based real-time motion artifact removal from wavelet subbands of ECG signal using particle filter [J], *Journal of Signal Processing Systems* 95 (8) (2023) 1021–1035.
- [26] M. Ma, M. Du, Q. Feng, et al., A new particle filter algorithm filtering motion artifact noise for clean electrocardiogram signals in wearable health monitoring system [J], *Rev. Sci. Instrum.* 95 (1) (2024).
- [27] C. Andrieu, A. Doucet, E. Punskeya, Sequential Monte Carlo methods for optimal filtering [M], *Sequential Monte Carlo methods in practice*. Springer. (2001) 79–95.
- [28] S. Banerjee, G.K. Singh, Monte Carlo filter-based motion artifact removal from electrocardiogram signal for real-time telecardiology system [J], *IEEE Trans. Instrum. Meas.* 70 (2021) 1–10.
- [29] H.D. Hesar, M. Mohebbi, An adaptive Kalman filter bank for ECG denoising [J], *IEEE J. Biomed. Health Inform.* 25 (1) (2020) 13–21.
- [30] BODILE R M, RAO T H. ECG Denoising Using Cubature Quadrature Kalman Filter Approach; proceedings of the 2020 IEEE India Council International Subsections Conference (INDISCON), F, 2020 [C]. IEEE.

- [31] H. Huang, S. Hu, Y. Sun, A discrete curvature estimation based low-distortion adaptive Savitzky–Golay filter for ECG denoising [J], *Sensors* 19 (7) (2019) 1617.
- [32] N.E. Huang, Z. Shen, S.R. Long, et al., The empirical mode decomposition and the Hilbert spectrum for nonlinear and non-stationary time series analysis [J], *Proceedings of the Royal Society of London Series a: Mathematical, Physical and Engineering Sciences* 1998 (454) (1971) 903–995.
- [33] D. Zhang, S. Wang, F. Li, et al., An efficient ECG denoising method based on empirical mode decomposition, sample entropy, and improved threshold function [J], *Wirel. Commun. Mob. Comput.* (2020) 1–11.
- [34] S.A. Malik, S.A. Parah, B.A. Malik, Power line noise and baseline wander removal from ECG signals using empirical mode decomposition and lifting wavelet transform technique [J], *Heal. Technol.* 12 (4) (2022) 745–756.
- [35] P.J. Luukko, J. Helske, E. Räsänen, Introducing libeemd: a program package for performing the ensemble empirical mode decomposition [J], *Comput. Stat.* 31 (2016) 545–557.
- [36] Y. Yue, C. Chen, X. Wu, et al., An effective electrocardiogram segments denoising method combined with ensemble empirical mode decomposition, empirical mode decomposition, and wavelet packet [J], *IET Signal Proc.* 17 (6) (2023) e12232.
- [37] SINGH P, SHAHNAWAZUDDIN S, PRADHAN G. Significance of modified empirical mode decomposition for ECG denoising; proceedings of the 2017 39th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC), F, 2017 [C]. IEEE.
- [38] M. Zhang, G. Wei, An integrated EMD adaptive threshold denoising method for reduction of noise in ECG [J], *PLoS One* 15 (7) (2020) e0235330.
- [39] M. Sraithi, Y. Jabrane, A denoising performance comparison based on ECG signal decomposition and local means filtering [J], *Biomed. Signal Process. Control* 69 (2021) 102903.
- [40] TORRES M E, COLOMINAS M A, SCHLOTTHAUER G, et al. A complete ensemble empirical mode decomposition with adaptive noise; proceedings of the 2011 IEEE international conference on acoustics, speech and signal processing (ICASSP), F, 2011 [C]. IEEE.
- [41] Y. Xu, M. Luo, T. Li, et al., ECG signal de-noising and baseline wander correction based on CEEMDAN and wavelet threshold [J], *Sensors* 17 (12) (2017) 2754.
- [42] R. Birok, ECG denoising using artificial neural networks and complete ensemble empirical mode decomposition [J], *Turkish Journal of Computer and Mathematics Education (TURCOMAT)* 12 (2) (2021) 2382–2389.
- [43] X. Xie, H. Liu, M. Shu, et al., A multi-stage denoising framework for ambulatory ECG signal based on domain knowledge and motion artifact detection [J], *Futur. Gener. Comput. Syst.* 116 (2021) 103–116.
- [44] U. Satija, B. Ramkumar, M.S. Manikandan, Automated ECG noise detection and classification system for unsupervised healthcare monitoring [J], *IEEE J. Biomed. Health Inform.* 22 (3) (2017) 722–732.
- [45] J. Moeyersons, E. Smets, J. Morales, et al., Artefact detection and quality assessment of ambulatory ECG signals [J], *Comput. Methods Programs Biomed.* 182 (2019) 105050.
- [46] K. Dragomiretskiy, D. Zosso, Variational mode decomposition [J], *IEEE Trans. Signal Process.* 62 (3) (2013) 531–544.
- [47] P. Singh, G. Pradhan, Variational mode decomposition based ECG denoising using non-local means and wavelet domain filtering [J], *Australas. Phys. Eng. Sci. Med.* 41 (2018) 891–904.
- [48] C. Li, Y. Wu, H. Lin, et al., ECG denoising method based on an improved VMD algorithm [J], *IEEE Sens. J.* 22 (23) (2022) 22725–22733.
- [49] P.G. Malghan, M.K. Hota, Grasshopper optimization algorithm based improved variational mode decomposition technique for muscle artifact removal in ECG using dynamic time warping [J], *Biomed. Signal Process. Control* 73 (2022) 103437.
- [50] H. Cao, L. Peyrodie, Variational mode decomposition-based simultaneous R peak detection and noise suppression for automatic ECG analysis [J], *IEEE Sens. J.* 23 (8) (2023) 8703–8713.
- [51] J. Mao, Z. Li, S. Li, et al., A novel ECG signal denoising algorithm based on sparrow search algorithm for optimal variational modal decomposition [J], *Entropy* 25 (5) (2023) 775.
- [52] V. Sirtoli, M. Lami, L.T. Lins, et al., Removal of motion artifacts in capacitive electrocardiogram acquisition: a review [J], *IEEE Trans. Biomed. Circuits Syst.* (2023).
- [53] C. Chen, M. Shu, S. Zhou, et al., Wavelet-domain group-sparse denoising method for ECG signals [J], *Biomed. Signal Process. Control* 83 (2023) 104702.
- [54] W. Jenkal, R. Latif, A. Tomanari, et al., An efficient algorithm of ECG signal denoising using the adaptive dual threshold filter and the discrete wavelet transform [J], *Biocybernetics and Biomedical Engineering* 36 (3) (2016) 499–508.
- [55] GON A, MUKHERJEE A. Removal of Noises from an ECG Signal Using an Adaptive S-Median Thresholding Technique; proceedings of the 2020 IEEE Applied Signal Processing Conference (ASPCON), F, 2020 [C]. IEEE.
- [56] T-N. Nguyen, T-H. Nguyen, V-T. Ngo, Artifact elimination in ECG signal using wavelet transform [J], *TELKOMNIKA (telecommunication Computing Electronics and Control)* 18 (2) (2020) 936–944.
- [57] WAN X-K, WU H, QIAO F, et al. Electrocardiogram baseline wander suppression based on the combination of morphological and wavelet transformation based filtering [J]. *Computational and mathematical methods in medicine*. 2019.
- [58] A. Kumar, H. Tomar, V.K. Mehla, et al., Stationary wavelet transform based ECG signal denoising method [J], *ISA Trans.* 114 (2021) 251–262.
- [59] NAGAI S, ANZAI D, WANG J. Motion artifact removal for wearable ECG using stationary wavelet multi-resolution analysis; proceedings of the 2017 IEEE 5th International Symposium on Electromagnetic Compatibility (EMC-Beijing), F, 2017 [C]. IEEE.
- [60] D. Berwal, C. Vandana, S. Dewan, et al., Motion artifact removal in ambulatory ECG signal for heart rate variability analysis [J], *IEEE Sens. J.* 19 (24) (2019) 12432–12442.
- [61] J. Gilles, Empirical wavelet transform [J], *IEEE Trans. Signal Process.* 61 (16) (2013) 3999–4010.
- [62] X. Xu, Y. Liang, P. He, et al., Adaptive motion artifact reduction based on empirical wavelet transform and wavelet thresholding for the non-contact ECG monitoring systems [J], *Sensors* 19 (13) (2019) 2916.
- [63] O. Singh, R.K. Sunkaria, ECG signal denoising via empirical wavelet transform [J], *Australas. Phys. Eng. Sci. Med.* 40 (2017) 219–229.
- [64] A.K. Verma, I. Saini, B.S. Saini, The baseline wandering noise removal from ECG signal using forward-backward Riemann Liouville fractional integral-based empirical wavelet transform approach [J], *Int. J. Wavelets Multiresolut. Inf. Process.* 16 (06) (2018) 1850049.
- [65] K. Yu, L. Feng, Y. Chen, et al., Accurate wavelet thresholding method for ECG signals [J], *Comput. Biol. Med.* 169 (2024) 107835.
- [66] C. Zou, Y. Qin, C. Sun, et al., Motion artifact removal based on periodical property for ECG monitoring with wearable systems [J], *Pervasive Mob. Comput.* 40 (2017) 267–278.
- [67] A. Agostinelli, A. Sbrillini, C. Giuliani, et al., Segmented beat modulation method for electrocardiogram estimation from noisy recordings [J], *Med. Eng. Phys.* 38 (6) (2016) 560–568.
- [68] S.K. Bashar, E. Ding, A.J. Walkey, et al., Noise detection in electrocardiogram signals for intensive care unit patients [J], *IEEE Access* 7 (2019) 88357–88368.
- [69] VINCENT P, LAROCHELLE H, BENGIO Y, et al. Extracting and composing robust features with denoising autoencoders; proceedings of the Proceedings of the 25th international conference on Machine learning, F, 2008 [C].
- [70] P. Aghaomidi, A. Mohammadisabab, J. Mazloun, et al., DeepRTSNet: deep robust two-stage networks for ECG denoising in practical use case [J], *IEEE Access* 10 (2022) 128232–128249.
- [71] ZUBAIR M, MOULI G N C, SHAIK R A. Removal of motion artifacts from ECG signals by combination of recurrent neural networks and deep neural networks; proceedings of the 2020 2nd International Conference on Electrical, Control and Instrumentation Engineering (ICECIE), F, 2020 [C]. IEEE.
- [72] E. Brophy, B. Hennelly, M. de Vos, et al., Improved electrode motion artefact denoising in ECG using convolutional neural networks and a custom loss function [J], *IEEE Access* 10 (2022) 54891–54898.
- [73] F. Samann, T. Schanze, RunDAE model: running denoising autoencoder models for denoising ECG signals [J], *Comput. Biol. Med.* 166 (2023) 107553.
- [74] A. Rasti-Meymandi, A. Ghaffari, A deep learning-based framework for ECG signal denoising based on stacked cardiac cycle tensor [J], *Biomed. Signal Process. Control* 71 (2022) 103275.
- [75] H-T. Chiang, Y-Y. Hsieh, S-W. Fu, et al., Noise reduction in ECG signals using fully convolutional denoising autoencoders [J], *IEEE Access* 7 (2019) 60806–60813.
- [76] L. Qiu, W. Cai, M. Zhang, et al., Two-stage ECG signal denoising based on deep convolutional network [J], *Physiol. Meas.* 42 (11) (2021) 115002.
- [77] W. Chorney, H. Wang, L. He, et al., Convolutional block attention autoencoder for denoising electrocardiograms [J], *Biomed. Signal Process. Control* 86 (2023) 105242.
- [78] P. Singh, A. Sharma, Attention-based convolutional denoising autoencoder for two-lead ECG denoising and arrhythmia classification [J], *IEEE Trans. Instrum. Meas.* 71 (2022) 1–10.
- [79] M. Chen, Y. Li, L. Zhang, et al., Elimination of random mixed noise in ECG using convolutional denoising autoencoder with transformer encoder [J], *IEEE J. Biomed. Health Inform.* (2024).
- [80] Y-S. Jhang, S-T. Wang, M-H. Sheu, et al., Integration design of portable ECG signal acquisition with deep-learning based electrode motion artifact removal on an embedded system [J], *IEEE Access* 10 (2022) 57555–57564.
- [81] Y-S. Jhang, S-T. Wang, M-H. Sheu, et al., Channel-Wise average pooling and 1D pixel-shuffle denoising autoencoder for electrode motion artifact removal in ECG [J], *Appl. Sci.* 12 (14) (2022) 6957.
- [82] L. Hu, W. Cai, Z. Chen, et al., A lightweight U-Net model for denoising and noise localization of ECG signals [J], *Biomed. Signal Process. Control* 88 (2024) 105504.
- [83] J. Wang, R. Li, R. Li, et al., Adversarial de-noising of electrocardiogram [J], *Neurocomputing* 349 (2019) 212–224.
- [84] F.L. Mvuh, A.C.O.V. Ebode ko', B. Bodo, Multichannel high noise level ECG denoising based on adversarial deep learning [J], *Sci. Rep.* 14 (1) (2024) 801.
- [85] P. Singh, G. Pradhan, A new ECG denoising framework using generative adversarial network [J], *IEEE/ACM Trans. Comput. Biol. Bioinf.* 18 (2) (2020) 759–764.
- [86] H. Wang, Y. Ma, A. Zhang, et al., Deep convolutional generative adversarial network with LSTM for ECG denoising [J], *Comput. Math. Methods Med.* (2023).
- [87] X. Wang, B. Chen, M. Zeng, et al., An ECG signal denoising method using conditional generative adversarial net [J], *IEEE J. Biomed. Health Inform.* 26 (7) (2022) 2929–2940.
- [88] P. Xiong, H. Wang, M. Liu, et al., ECG signal enhancement based on improved denoising auto-encoder [J], *Eng. Appl. Artif. Intel.* 52 (2016) 194–202.
- [89] Y. Hou, R. Liu, M. Shu, et al., An ECG denoising method based on adversarial denoising convolutional neural network [J], *Biomed. Signal Process. Control* 84 (2023) 104964.
- [90] F.P. Romero, D.C. Piñol, C.R. Vázquez-SEISDEDOS, DeepFilter: an ECG baseline wander removal filter using deep learning techniques [J], *Biomed. Signal Process. Control* 70 (2021) 102992.

- [91] H. Li, G. Ditzler, J. Roveda, et al., DeScoD-ECG: deep score-based diffusion model for ECG baseline wander and noise removal [J], *IEEE J. Biomed. Health Inform.* (2023).
- [92] Y. Hou, R. Liu, M. Shu, et al., Deep neural network denoising model based on sparse representation algorithm for ECG signal [J], *IEEE Trans. Instrum. Meas.* 72 (2023) 1–11.
- [93] Y. Jin, C. Qin, J. Liu, et al., A novel deep wavelet convolutional neural network for actual ecg signal denoising [J], *Biomed. Signal Process. Control* 87 (2024) 105480.
- [94] M-H. Sheu, Y-S. Jhang, Y-C. Chang, et al., Lightweight denoising autoencoder design for noise removal in electrocardiography [J], *IEEE Access* 10 (2022) 98104–98116.
- [95] G.B. Moody, W. Muldrow, R.G. Mark, A noise stress test for arrhythmia detectors [J], *Comput. Cardiol.* 11 (3) (1984) 381–384.
- [96] MANIKANDAN M S, DANDAPAT S. ECG distortion measures and their effectiveness; proceedings of the 2008 First International Conference on Emerging Trends in Engineering and Technology, F, 2008 [C]. IEEE.
- [97] E. Dasan, I. Panneerselvam, A novel dimensionality reduction approach for ECG signal via convolutional denoising autoencoder with LSTM [J], *Biomed. Signal Process. Control* 63 (2021) 102225.
- [98] ANTICZAK K. Deep recurrent neural networks for ECG signal denoising [J]. arXiv preprint arXiv:1807.11551, 2018.
- [99] J.T. Gwin, K. Gramann, S. Makeig, et al., Removal of movement artifact from high-density EEG recorded during walking and running [J], *J. Neurophysiol.* 103 (6) (2010) 3526–3534.
- [100] KIM B H, CHUN J, JO S. Dynamic motion artifact removal using inertial sensors for mobile BCI; proceedings of the 2015 7th International IEEE/EMBS Conference on Neural Engineering (NER), F, 2015 [C]. IEEE.
- [101] O. Friman, M. Borga, P. Lundberg, et al., Exploratory fMRI analysis by autocorrelation maximization [J], *Neuroimage* 16 (2) (2002) 454–464.
- [102] P.K. Shukla, V. Roy, P.K. Shukla, et al., An advanced EEG motion artifacts eradication algorithm [J], *Comput. J.* 66 (2) (2023) 429–440.
- [103] R. Ranjan, B.C. Sahana, A.K. Bhandari, Motion artifacts suppression from EEG signals using an adaptive signal denoising method [J], *IEEE Trans. Instrum. Meas.* 71 (2022) 1–10.
- [104] L. Neri, M.T. Oberdier, K.C. van Abeelen, et al., Electrocardiogram monitoring wearable devices and artificial-intelligence-enabled diagnostic capabilities: a review [J], *Sensors* 23 (10) (2023) 4805.
- [105] M. Bodini, M.W. Rivolta, R. Sassi, Opening the black box: interpretability of machine learning algorithms in electrocardiography [J], *Phil. Trans. R. Soc. A* 379 (2212) (2021) 20200253.
- [106] D. Zhang, S. Yang, X. Yuan, et al., Interpretable deep learning for automatic diagnosis of 12-lead electrocardiogram [J], *Iscience* 24 (4) (2021).
- [107] C. Han, J. Sun, Y. Bian, et al., Automated detection and localization of myocardial infarction with interpretability analysis based on deep learning [J], *IEEE Trans. Instrum. Meas.* 72 (2023) 1–12.



Juya Zhang received the B.E. degree from Shandong Youth University Of Political Science, Jinan, China, in 2022. She is presently pursuing the M.S. degree in Tiangong University, Tianjin, China. Her research interests include ECG signal noise reduction and biomedical signal processing and analysis.



Yu Guo received the B.E. degree from Eastern Liaoning University, Dandong, China, in 2018. He is currently working in the School of Life Sciences at Tiangong University, Tianjin, China. His main research interests include Wearable medical device development, biomedical signal processing and analysis, vital signs monitoring and life support technology.



Xinming Dong graduated from Air Force Medical College in 1990, majoring in Medical Instrumentation, and joined the Air Force Tianjin Hospital in the same year, and graduated from Air Force Political Science College in 1996, majoring in Biomedical Engineering, and is now the deputy director and deputy chief technician of Tianjin Rehabilitation and Convalescent Center. He has been engaged in biomedical engineering since then, and is now the deputy director of the convalescent department of Tianjin Rehabilitation and Recuperation Center, and deputy chief technician.



Tong Wang received the B.E. degree from Zhengzhou Institute of Industrial Application Technology, Zhengzhou, China, in 2021. He is presently pursuing the M.S. degree in Tiangong University, Tianjin, China. His main research interests include physiological parameter acquisition and environmental applicability of equipment.



Jinhai Wang received the B.E. degree from Information and Control Engineering Department, Xian Jiaotong University, China in 1987, and his PhD degree from Precision Instrument and Opto-Electronics Engineering Department, Tianjin University, China in 2007, worked for Tiangong University in 1987. He then did research work as a senior visiting scholar in University of Auckland, New Zealand in 2005. Dr. Wang then joined the faculty team in Tiangong University in 2006 as a professor, and has Engaged in teaching and research work in Medical Electronics for a long time for the same university. He is also very active in serving as an Executive director of Tianjin Society of Biomedical Engineering, and vice director of Tianjin Society for Intelligent Science and Technology.



Xin Ma received his B.A. and M.A. in Computer Science and obtained his Ph.D. in Textile Science and Engineering from Tiangong University, China. From 2018 to 2019, he was a visiting scholar at Department of Bioengineering, UCLA. His research interest is the combination of computer science and biomedical engineering to improve human health and medical treatments through biointegrated and bioinspired design innovation.



Huiquan Wang received the B.E. degree from Biomedical Engineering of Tianjin University, China in 2008, and his PhD degree from Biomedical Engineering, Tianjin University, China in 2014. He worked for Tiangong University in 2024. He did research work as a senior visiting scholar worked in Johns Hopkins University in 2005. Dr. Wang then joined the faculty team in Tiangong University in 2014 as a professor, and has engaged in teaching and research work in Medical Electronics for a long time for the same university. He is also very active in serving as an vice director of Tianjin Society for Intelligent Science and Technology.